Open Yale Courses
Yale University

PHYS 200
Fundamentals of Physics I

Instructor:
Prof. Ramamurti Shankar

Notes & Summary Prepared By:
Samah Shehata

Course Topics:
Newtonian Mechanics · Special Relativity · Gravitation
Thermodynamics · Waves

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1 Introduction to Newtonian Mechanics

Newtonian Mechanics forms the foundation of classical physics and deals with the motion of objects and the forces that cause these motions. It is divided into two main branches: **Kinematics** and **Dynamics**.

1.1 Kinematics

Kinematics is the study of *motion itself*, without considering its causes. It focuses on describing how objects move using measurable quantities such as:

- **Position** :the location of an object in space.
- **Velocity** : the rate of change of position with respect to time.
- **Acceleration** :the rate of change of velocity with respect to time.

In other words, kinematics describes *what happens* during motion, not *why* it happens.

> Example: Observing a car move along a straight road and measuring its speed, displacement, and acceleration without analyzing the engine or the forces acting on it is a study in kinematics.

1.2 Dynamics

Dynamics, on the other hand, deals with the *causes of motion*. It connects motion to the forces acting on an object and its mass, primarily through Newton's laws of motion.

> Example: Explaining that the car accelerates because the engine exerts a forward force and friction resists the motion that is dynamics.

1.3 Summary

Kinematics	**Dynamics**
Describes motion (what happens)	Explains motion (why it happens)
No concern with forces	Involves forces and Newton's laws
Quantities: position, velocity, acceleration	Quantities: mass, force, momentum, energy

2 Models of a Physical Body

Before studying motion, it is essential to define the kind of object (or body) being analyzed. In mechanics, different models are used to represent a physical body depending on the level of detail required.

2.1 Particle (or Point Mass)

A **particle** is an idealized model of a body that has **mass but no size or shape**. All of its mass is considered to be concentrated at a single point in space.

- It has no dimensions you cannot distinguish its rotation or orientation.
- Motion is described only by changes in its position.
- Suitable for studying objects whose size is negligible compared to the distances they move.

  Example: Treating the Earth as a point mass when studying its orbit around the Sun.

2.2 Rigid Body

A **rigid body** is composed of many particles, but the distance between any two particles remains **constant** at all times. It has definite **size and shape**, and it can undergo both **translation** and **rotation**.

Example: A spinning wheel or a rotating planet can be modeled as rigid bodies.

2.3 Deformable Body

In a **deformable body**, the distances between its particles can change meaning it can stretch, compress, or bend. This model is used in more advanced studies such as elasticity and fluid mechanics.

Example: A rubber ball or a piece of metal under stress.

2.4 Comparison of Models

Type of Body	**Has Size/Shape?**	**Can Rotate?**	**Can Deform?**	**Example**
Particle (Point Mass)	No	No	No	Planet orbiting the Sun
Rigid Body	Yes	Yes	No	Rotating wheel
Deformable Body	Yes	Yes	Yes	Rubber ball, metal rod

3 Functions and Graphical Representation in Physics

Mathematics provides the language through which physics describes motion. A key concept in this language is the **function**, which expresses how one quantity depends on another.

3.1 Meaning of a Function

A function defines a relationship between two variables. For example, in motion:

$$x = f(t)$$

the position x depends on time t.

Sometimes we write the function with a name, such as:

$$f(t) = t^2,$$

and sometimes we write it directly as:

$$x(t) = t^2.$$

Both express how x changes as time passes.

3.2 Importance of Graphs in Physics

In physics, we often use graphs to visualize how quantities vary with time or with one another. When observing a graph, it is essential to know **what physical information** it conveys.

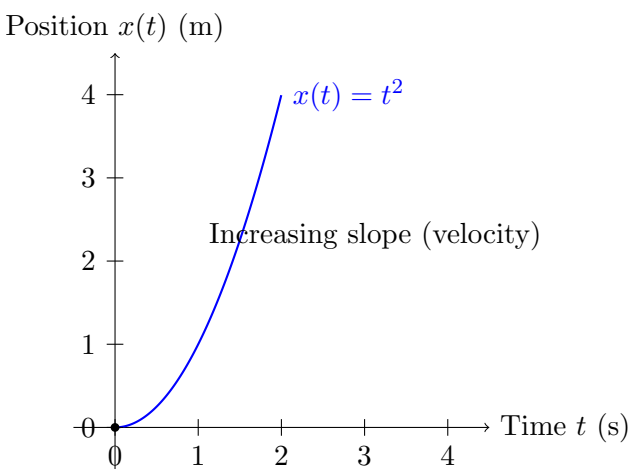
What to Look At	What It Tells You
Slope of the graph	Indicates the rate of change. For $x(t)$, the slope gives velocity.
Curvature (bending)	Shows whether the rate of change is itself changing i.e., acceleration.
Intercepts with axes	Provide initial conditions, such as starting position or time.
Units on axes	Identify what physical quantities are compared.
Shape of the curve	Distinguishes uniform motion (straight line) from accelerated motion (curve).

3.3 Example: The Function $x(t) = t^2$

For the function $x(t) = t^2$:

- The graph is a **parabola** opening upward.
- The **slope** (velocity) increases with time.
- The motion represents an **accelerating object**.

  Physically, this could describe an object starting from rest and speeding up as it moves, such as a ball rolling down an inclined plane.



4 Instantaneous and Average Velocity

When describing motion, velocity tells us how fast and in which direction an object's position changes with time.

4.1 Average Velocity

The **average velocity** of an object over a time interval is defined as:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

It represents the overall change in position divided by the time it took.

Example: If a car moves from $x_1 = 0\,\text{m}$ to $x_2 = 100\,\text{m}$ in 5 seconds,

$$v_{\text{avg}} = \frac{100 - 0}{5} = 20\,\text{m/s}.$$

The average velocity is the **slope of the secant line** connecting two points on the position–time graph.

4.2 Instantaneous Velocity

The **instantaneous velocity** is the velocity at a specific instant of time. It is defined as the limit of the average velocity as the time interval becomes infinitesimally small:

$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}.$$

This is the derivative of position with respect to time, and it represents the **slope of the tangent line** to the curve $x(t)$ at a given point.

Example: If $x(t) = t^2$, then

$$v = \frac{dx}{dt} = 2t.$$

At $t = 1\,\text{s}$, $v = 2\,\text{m/s}$; At $t = 2\,\text{s}$, $v = 4\,\text{m/s}$.

4.3 Numerical Value of Velocity

When we substitute a specific time value into the velocity equation, we get the **numerical value** of velocity the computed number with its unit. For example:

$$v = 2t \quad \Rightarrow \quad v(3) = 6\,\text{m/s}.$$

So, the **instantaneous velocity** is the concept (the derivative), and its **numerical value** is the calculated number for a chosen moment.

In summary, the instantaneous velocity represents the "speed at this exact moment," and the numerical value tells us its actual magnitude at that moment.

5 Integration in Motion: The Role of Constants and Initial Conditions

In physics, acceleration is often known or measured, and our goal is to find how velocity and position change with time. To move from acceleration to velocity and position, we use **integration** the mathematical opposite of differentiation.

5.1 From Acceleration to Velocity

Starting from the definition:

$$a = \frac{dv}{dt}$$

we can obtain velocity by integrating both sides with respect to time:

$$v = \int a\,dt.$$

However, unlike differentiation, integration is **not a unique process**. It always introduces an **arbitrary constant** because many different functions can have the same derivative.

> "Integration is like guessing the opposite of differentiation. You can always check your guess by differentiating."

So the result becomes:

$$v = at + C_1$$

where C_1 is a constant.

5.2 Physical Meaning of the Constant

Mathematically, there is no issue with adding any constant, but in **physics**, every constant must carry a **physical meaning**.

We determine C_1 by using an **initial condition** for example, when $t = 0$, the velocity is v_0:

$$v(0) = v_0 = a(0) \times 0 + C_1 \Rightarrow C_1 = v_0.$$

Therefore, the complete physical expression for velocity is:

$$\boxed{v = v_0 + at}$$

Here:

- a: acceleration (assumed constant)
- v_0: initial velocity at $t = 0$

5.3 From Velocity to Position

To obtain position, we integrate the velocity function:

$$x = \int v\,dt = \int (v_0 + at)\,dt = v_0 t + \frac{1}{2}at^2 + C_2$$

where C_2 is another integration constant.

Again, this constant represents an **initial condition** the position of the particle when $t = 0$:

$$x(0) = x_0 = v_0(0) + \frac{1}{2}a(0)^2 + C_2 \Rightarrow C_2 = x_0.$$

Thus, the complete position equation becomes:

$$\boxed{x = x_0 + v_0 t + \frac{1}{2}at^2}$$

5.4 Why the Constants Matter

Pure mathematics allows us to add constants freely, but physics demands we *interpret* them.

- x_0 tells us **where** the object started.
- v_0 tells us **how fast** it was moving initially.
- a tells us **how its motion changes**.

Without x_0 and v_0, acceleration alone cannot describe the full motion we would know the rate of change, but not the actual location of the particle at any instant.

> "In mathematics, adding a constant changes nothing. In physics, it defines the story of where motion begins."

5.5 Summary of Kinematic Equations for Constant Acceleration

Quantity	Equation (for constant acceleration)
Velocity-time relation	$v = v_0 + at$
Position-time relation	$x = x_0 + v_0 t + \frac{1}{2}at^2$
Velocity-position relation	$v^2 = v_0^2 + 2a(x - x_0)$

These three equations describe the motion of a particle under constant acceleration completely they contain all the information about its velocity, position, and change over time.

6 Dimensions in Mechanics

In classical mechanics, physical motion is described within one, two, or three spatial dimensions.

- **1D Motion:** Movement occurs along a straight line, such as a particle sliding on a track.
- **2D Motion:** Motion is confined to a plane, as in projectile motion or circular paths.
- **3D Motion:** The most general case, describing motion in physical space (x, y, z).

Beyond these familiar dimensions, modern theoretical physics introduces the idea that reality may extend into higher-dimensional spaces. String theory, for instance, postulates that fundamental particles are not point-like but one-dimensional vibrating strings existing within a ten-dimensional spacetime nine spatial dimensions and one temporal dimension.

This framework unifies the principles of quantum mechanics and general relativity by suggesting that different vibrational modes of strings correspond to different particles, and that the extra dimensions are compactified or "curled up" at scales beyond current experimental reach.

7 Introduction to Vectors

To describe motion in physical space, we need quantities that carry both *magnitude* and *direction*. Such quantities are called **vectors**.

Consider a simple example: imagine going on a camping trip. On the first day, you walk 5 km. On the second day, you walk another 5 km. If you are asked, "Where are you now relative to the camp?", you cannot answer this question without knowing the *direction* of your movement.

Even if you restricted yourself to moving along a single axis, say the x-axis, you still need to specify whether you went to the right or to the left. Thus, in one dimension, direction can be captured by a *sign* positive or negative. A displacement of $+5$ km indicates movement to the right, while -5 km indicates movement to the left.

However, in two or three dimensions, direction cannot be represented by a sign alone, since there are infinitely many possible directions in which one can move. For such cases, we introduce the concept of a **vector displacement**.

Let us define two displacements:

$$\vec{A} \quad \text{(Day 1 displacement)}, \quad \vec{B} \quad \text{(Day 2 displacement)}.$$

Each vector has:

- a **magnitude** (the length of the arrow), representing the distance moved,
- and a **direction** (the orientation of the arrow), representing the path taken.

Graphically, vectors are represented by arrows drawn from a starting point to an endpoint. For example, $\vec{A}$ may start from the origin and end at a point (x_1, y_1).

The rule for adding two vectors follows naturally from physical intuition: if you perform displacement $\vec{A}$ followed by displacement $\vec{B}$, the resultant displacement $\vec{C}$ (from your starting point to your final position) is given by

$$\vec{C} = \vec{A} + \vec{B}.$$

This geometrical addition represents the total effect of successive motions and forms the basis of vector algebra used throughout mechanics.

8 Vectors

Vectors are central to the mathematical description of physics because they carry both a *magnitude* and a *direction*. This section introduces notation, algebra, and the most important vector operations used in mechanics.

8.1 Definition and Notation

A **vector** is denoted by an arrow above the symbol, e.g. $\vec{A}$, or in boldface $\mathbf{A}$. A plain letter A denotes a *scalar* (a number). The magnitude (length) of $\vec{A}$ is written $|\vec{A}|$ or A when the meaning is clear. The unit vector that points in the direction of $\vec{A}$ is

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}.$$

8.2 Components and Unit Vectors

In a Cartesian coordinate system in two or three dimensions, any vector may be expressed as a linear combination of orthonormal basis vectors. In 3D it is conventional to use

$$\hat{\imath},\ \hat{\jmath},\ \hat{k}$$

so that

$$\vec{A} = A_x\,\hat{\imath} + A_y\,\hat{\jmath} + A_z\,\hat{k},$$

where the components are

$$A_x = \vec{A}\cdot\hat{\imath}, \qquad A_y = \vec{A}\cdot\hat{\jmath}, \qquad A_z = \vec{A}\cdot\hat{k}.$$

In 2D one drops the z-component:

$$\vec{A} = A_x\,\hat{\imath} + A_y\,\hat{\jmath}.$$

8.3 Geometric (Tip-to-Tail) Addition and Component-wise Addition

Geometrically, if a particle undergoes displacement $\vec{A}$ and then $\vec{B}$, the resultant displacement is

$$\vec{C} = \vec{A} + \vec{B},$$

found by placing the tail of $\vec{B}$ at the tip of $\vec{A}$ (the *tip-to-tail* rule). Vector addition is commutative and associative:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}, \qquad (\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C}).$$

Component-wise, addition is performed by adding corresponding components:

$$\vec{A} + \vec{B} = (A_x + B_x)\,\hat{\imath} + (A_y + B_y)\,\hat{\jmath} + (A_z + B_z)\,\hat{k}.$$

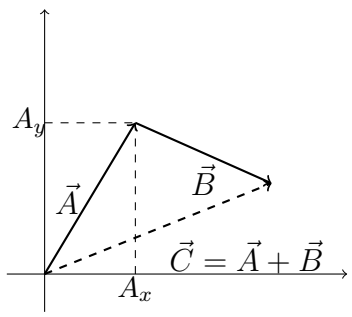


Figure 1: Tip-to-tail addition and component decomposition in the plane.

8.4 Scalar Multiplication

Multiplying a vector by a scalar λ scales its magnitude and (if $\lambda < 0$) reverses its direction:

$$\lambda\vec{A} = (\lambda A_x)\,\hat{\imath} + (\lambda A_y)\,\hat{\jmath} + (\lambda A_z)\,\hat{k}.$$

8.5 Dot (Scalar) Product

The dot product of two vectors is defined algebraically and geometrically by

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z = |\vec{A}|\,|\vec{B}| \cos\theta,$$

where θ is the angle between $\vec{A}$ and $\vec{B}$. Key properties:

- Commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$.
- Distributive: $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$.
- $\vec{A} \cdot \vec{A} = |\vec{A}|^2$ (gives the squared magnitude).

The dot product is widely used to compute projections and work done by a force: if a constant force $\vec{F}$ acts over a displacement $\vec{d}$, the work is $W = \vec{F} \cdot \vec{d}$.

8.6 Cross (Vector) Product (3D only)

The cross product is defined for three-dimensional vectors and produces a vector perpendicular to both operands. Its magnitude and direction are given by

$$|\vec{A} \times \vec{B}| = |\vec{A}|\,|\vec{B}| \sin\theta, \qquad \hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|},$$

where θ is the angle between $\vec{A}$ and $\vec{B}$ and the direction follows the right-hand rule. Component form (determinant notation):

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}.$$

Properties:

- Anti-commutative: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$.
- Distributive: $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$.
- $\vec{A} \times \vec{A} = \vec{0}$.

8.7 Useful Identities

$$\begin{aligned} \vec{A} \cdot (\vec{B} + \vec{C}) &= \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}, \\ \vec{A} \times (\vec{B} + \vec{C}) &= \vec{A} \times \vec{B} + \vec{A} \times \vec{C}, \\ (\alpha\vec{A}) \times (\beta\vec{B}) &= \alpha\beta\,(\vec{A} \times \vec{B}). \end{aligned}$$

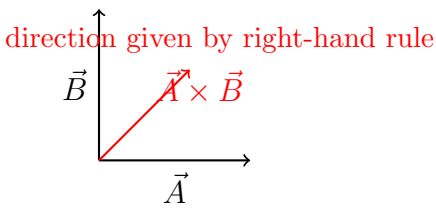


Figure 2: Schematic of $\vec{A} \times \vec{B}$ (direction shown schematically).

8.8 Examples

1. Addition and components. Let $\vec{A} = 3\,\hat{\imath} + 2\,\hat{\jmath}$ and $\vec{B} = -1\,\hat{\imath} + 4\,\hat{\jmath}$. Then

$$\vec{A} + \vec{B} = (3 + (-1))\,\hat{\imath} + (2 + 4)\,\hat{\jmath} = 2\,\hat{\imath} + 6\,\hat{\jmath}.$$

2. Dot product. Using the same vectors,

$$\vec{A} \cdot \vec{B} = 3(-1) + 2(4) = -3 + 8 = 5.$$

3. Cross product (3D). Let $\vec{A} = \hat{\imath} + 2\,\hat{\jmath} + 0\,\hat{k}$ and $\vec{B} = 2\,\hat{\imath} - \hat{\jmath} + \hat{k}$. Then compute the determinant to find $\vec{A} \times \vec{B}$

Compute the cross product using the determinant form:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 2 & 0 \\ 2 & -1 & 1 \end{vmatrix} = \hat{\imath}(2 \cdot 1 - 0 \cdot (-1)) - \hat{\jmath}(1 \cdot 1 - 0 \cdot 2) + \hat{k}(1 \cdot (-1) - 2 \cdot 2).$$

Therefore

$$\vec{A} \times \vec{B} = 2\hat{\imath} - 1\hat{\jmath} - 5\hat{k} = \big(2,\ -1,\ -5\big).$$

Verification. The cross product should be perpendicular to both $\vec{A}$ and $\vec{B}$. Check the dot products:

$$\vec{A} \cdot (\vec{A} \times \vec{B}) = 1 \cdot 2 + 2 \cdot (-1) + 0 \cdot (-5) = 2 - 2 + 0 = 0,$$

$$\vec{B} \cdot (\vec{A} \times \vec{B}) = 2 \cdot 2 + (-1) \cdot (-1) + 1 \cdot (-5) = 4 + 1 - 5 = 0.$$

Both are zero, confirming $\vec{A} \times \vec{B}$ is orthogonal to $\vec{A}$ and $\vec{B}$.

8.9 Physical interpretation of vector operations

Vector addition. The sum $\vec{C} = \vec{A} + \vec{B}$ represents the resultant effect of two vector actions. Physically, if $\vec{A}$ and $\vec{B}$ are displacements then $\vec{C}$ is the total displacement; if they are forces then $\vec{C}$ is the net force that determines the acceleration via $\vec{F}_{\text{net}} = m\vec{a}$.

Scalar multiplication. Scaling $\vec{A}$ by λ changes its magnitude (and reverses its direction if $\lambda < 0$). This models increasing/decreasing the strength of a force or reversing a velocity.

Dot product.

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta$$

The dot product returns the component of one vector along the other. Physically:

$$W = \vec{F} \cdot \vec{d}$$

gives the work done only the force component parallel to the displacement contributes.

Cross product (3D).

$$\vec{A} \times \vec{B}$$

has magnitude $|\vec{A}||\vec{B}| \sin\theta$ and direction given by the right-hand rule. Physically:

$$\vec{\tau} = \vec{r} \times \vec{F} \quad \text{(torque)}, \qquad \vec{L} = \vec{r} \times \vec{p} \quad \text{(angular momentum)},$$

both describe rotational effects where perpendicular components produce rotation.

Scalar triple product.

$$(\vec{A} \times \vec{B}) \cdot \vec{C}$$

equals the signed volume of the parallelepiped formed by $\vec{A}, \vec{B}, \vec{C}$, used in flux/volume calculations.

Component resolution. Writing $\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$ lets us apply Newton's laws per coordinate this is essential for solving dynamics problems (equilibrium, inclined planes, etc.).

Physical example. Let $\vec{F}_1 = (3, 0)$ N and $\vec{F}_2 = (0, 4)$ N:

$$\vec{F}_{\text{net}} = (3, 4) \text{ N}, \quad |\vec{F}_{\text{net}}| = 5 \text{ N}.$$

If $\vec{d} = (2, 0)$ m, then $W = \vec{F}_{\text{net}} \cdot \vec{d} = 6$ J (only the x-component contributes).

8.10 Rotation of a Vector and Basis Transformation

A vector $\vec{A}$ is an entity that has a physical meaning independent of the coordinate system used to represent it. Therefore, **the same vector can have two equivalent representations** in two different coordinate bases: the unprimed basis $(\hat{\imath}, \hat{\jmath})$ and the primed basis $(\hat{\imath}', \hat{\jmath}')$.

$$\vec{A} = A_x\,\hat{\imath} + A_y\,\hat{\jmath} = A'_x\,\hat{\imath}' + A'_y\,\hat{\jmath}'.$$

Here, A_x, A_y and A'_x, A'_y are the components of the same vector expressed in the two coordinate systems. The vector itself does not change only its numerical components do, depending on the orientation of the coordinate axes.

If the primed coordinate system is rotated counterclockwise by an angle θ with respect to the unprimed system, then the basis vectors are related by:

$$\begin{aligned}\hat{\imath}' &= \cos\theta\,\hat{\imath} + \sin\theta\,\hat{\jmath},\\ \hat{\jmath}' &= -\sin\theta\,\hat{\imath} + \cos\theta\,\hat{\jmath}.\end{aligned}$$

Substituting these into the expression for $\vec{A}$, we can relate the components in both systems:

$$\begin{pmatrix} A'_x \\ A'_y \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}.$$

Alternatively, this can be inverted to get:

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} A'_x \\ A'_y \end{pmatrix}.$$

This transformation shows how the vector components change when the coordinate axes rotate, while the vector itself remains invariant. Such relationships form the foundation for understanding rotations, transformations, and tensor operations in higher-level physics.

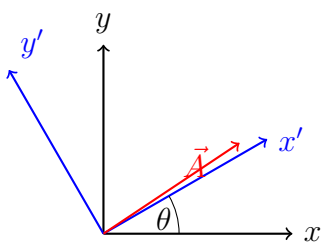


Figure 3: Rotation of a vector's components between two coordinate systems.

Inverting (or using the transpose of the rotation matrix) yields the primed components in terms of the unprimed components:

$$\boxed{A'_x = A_x\cos\phi + A_y\sin\phi, \qquad A'_y = -A_x\sin\phi + A_y\cos\phi}.$$

Physical meaning. The geometric vector $\vec{A}$ is unchanged by a rotation of the coordinate axes; only its numerical description (components) changes. Rotation matrices preserve length and angles, so

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2} = \sqrt{{A'_x}^2 + {A'_y}^2}.$$

Numeric check (example). For $(A_x, A_y) = (3, 4)$ and $\phi = 30^\circ$ ($\cos 30^\circ \approx 0.8660$, $\sin 30^\circ = 0.5$) one finds

$$A'_x = 3 \cdot 0.866 + 4 \cdot 0.5 \approx 4.598, \qquad A'_y = -3 \cdot 0.5 + 4 \cdot 0.866 \approx 1.964,$$

and $\sqrt{4.598^2 + 1.964^2} \approx 5$, matching the original length $\sqrt{3^2 + 4^2} = 5$.

Vectors are central to the mathematical description of physics because they carry both a *magnitude* and a *direction*. This section introduces notation, algebra, and the most important vector operations used in mechanics.

9 Arrow vs Entity in Vectors

9.1 Arrow (Geometric Picture)

When we draw a vector as an arrow say $\vec{A}$ we're showing its geometric nature:

- **Direction** $\rightarrow$ the way the arrow points.
- **Magnitude (length)** $\rightarrow$ how strong or large the vector is.

This arrow is an object that exists in space. It doesn't depend on how we describe it it just "is."

For example:

- A displacement from one point to another.
- A force acting on an object.
- A velocity of a moving particle.

These are geometric arrows in the real world; they exist regardless of what coordinate axes we choose.

$\Rightarrow$ The arrow is the same for every observer it represents a real, physical entity in space.

9.2 Entity (Mathematical Representation)

When we write something like:

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath},$$

we're representing that arrow using a set of numbers (components) multiplied by basis vectors (unit vectors along the axes).

Here, $\vec{A}$ is a **mathematical entity** it lives in an abstract vector space. It can be expressed in different bases (unprimed, primed, rotated, etc.), but the entity itself doesn't change.

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} = A'_x\hat{\imath}' + A'_y\hat{\jmath}'.$$

Both sides describe the same entity (the same arrow), but in different coordinate systems.

9.3 In Simple Words

Concept	**Represents**
Arrow (Geometric Vector)	The physical direction and magnitude (the "thing" itself)
Entity (Mathematical Representation)	The components and basis we use to describe the arrow

9.4 In Physics Language

- The arrow = the physical quantity (like force, velocity, or displacement).
- The entity = the symbolic or numerical representation of that quantity in a chosen coordinate frame.

So we might say:

"The vector $\vec{A}$ is an entity that represents a geometric arrow in space."

10 The Concept of Limit and Its Role

In physics and calculus, the concept of a **limit** allows us to describe how a quantity behaves as another quantity approaches a specific value. This idea becomes essential when we want to move from describing *average change* to describing *instantaneous change*.

10.1 Average Velocity

If a particle moves along a line and its position at time t is $x(t)$, then the **average velocity** over a finite time interval Δt is

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{x(t + \Delta t) - x(t)}{\Delta t}.$$

This ratio tells us how much the position changes per unit time, but it only gives an *average rate of change* over the interval Δt not the exact velocity at any single instant.

10.2 The Limiting ProcessMoving Toward the Instantaneous Value

To find the velocity *at an instant* (say, exactly at time t), we must make the time interval Δt infinitely small so that we capture the change *right at that moment.* This is done mathematically by taking the **limit**:

$$v(t) = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}.$$

This limit defines the **instantaneous velocity**, and the same idea defines the derivative of $x(t)$:

$$v(t) = \frac{dx}{dt}.$$

10.3 Why the Limit is Necessary

The limit bridges the gap between discrete change and continuous motion:

- Without the limit, we only know how the position changes *over* an interval.
- With the limit, we find how the position changes *at a specific instant.*

Physically, this means that the particle's motion is smooth and continuous, not jumping between pointsso by taking Δt smaller and smaller, we approach the *true velocity* at that moment.

10.4 Example:

Suppose the displacement of a particle is given by

$$x(t) = 4t^2.$$

The average velocity between t and $t + \Delta t$ is

$$v_{\text{avg}} = \frac{x(t + \Delta t) - x(t)}{\Delta t} = \frac{4(t + \Delta t)^2 - 4t^2}{\Delta t}.$$

Simplifying,

$$v_{\text{avg}} = \frac{4(t^2 + 2t\Delta t + (\Delta t)^2) - 4t^2}{\Delta t} = \frac{8t\Delta t + 4(\Delta t)^2}{\Delta t} = 8t + 4\Delta t.$$

Now, take the limit as $\Delta t \to 0$:

$$v(t) = \lim_{\Delta t \to 0} (8t + 4\Delta t) = 8t.$$

Thus, the instantaneous velocity is $v(t) = 8t$.

10.5 Physical Meaning

The limit operation $\lim_{\Delta t \to 0}$ zooms in on the motion so closely that we can determine the exact rate of change of position the instantaneous velocity at each moment in time.

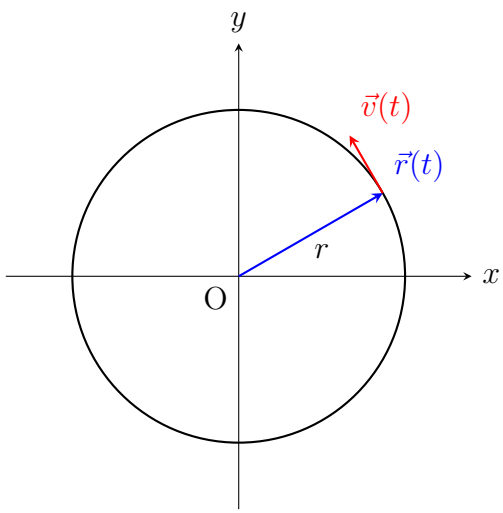
Geometrically, this process converts the **secant line** (which connects two points on the $x(t)$ curve) into the **tangent line** at a single point, representing the slope of the curve at that instant.

In Words: The limit extracts the true, instantaneous behavior of a changing quantity. When applied to motion, it transforms the average velocity (change over a time interval) into the instantaneous velocity (change at a specific instant). It is the mathematical lens through which we view continuous physical change.

11 Uniform Circular Motion

11.1 Introduction

A particle is said to execute **uniform circular motion** when it moves in a circle of fixed radius r with constant speed v. Even though the magnitude of its velocity remains constant, its *direction* changes continuously. Because of this change in direction, the particle experiences an **acceleration** toward the center of the circle the *centripetal acceleration.*



11.2 Position Vector

At any instant t, the position of the particle can be represented by the vector

$$\vec{r}(t) = r\cos(\omega t)\,\hat{\imath} + r\sin(\omega t)\,\hat{\jmath},$$

where:

- r is the radius of the circular path,
- ω is the angular velocity (rate of change of angle θ with respect to time, i.e. $\omega = \frac{d\theta}{dt}$),
- $(\hat{\imath}, \hat{\jmath})$ are the unit vectors along the x and y axes.

This expression shows that as time passes, the components of $\vec{r}(t)$ oscillate sinusoidally meaning the particle traces a circle of radius r centered at the origin.

11.3 Velocity Vector Derivative of Position

The **velocity vector** is the time derivative of the position vector:

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt}.$$

Differentiating component-wise gives:

$$\begin{aligned}\vec{v}(t) &= \frac{d}{dt}\left[r\cos(\omega t)\,\hat{\imath} + r\sin(\omega t)\,\hat{\jmath}\right] \\ &= r(-\omega\sin(\omega t))\,\hat{\imath} + r(\omega\cos(\omega t))\,\hat{\jmath}.\end{aligned}$$

Thus,

$$\boxed{\vec{v}(t) = r\omega[-\sin(\omega t)\,\hat{\imath} + \cos(\omega t)\,\hat{\jmath}]}.$$

The magnitude of $\vec{v}$ is constant:

$$|\vec{v}| = \sqrt{(-r\omega\sin(\omega t))^2 + (r\omega\cos(\omega t))^2} = r\omega = v.$$

Hence the *speed* v is constant, but $\vec{v}$ changes direction continuously always tangent to the circular path.

11.4 Acceleration Vector Derivative of Velocity

The acceleration vector is the derivative of the velocity vector:

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt}.$$

Differentiating,

$$\begin{aligned}\vec{a}(t) &= \frac{d}{dt}\big[r\omega(-\sin(\omega t)\,\hat{\imath} + \cos(\omega t)\,\hat{\jmath})\big] \\ &= r\omega[-\omega\cos(\omega t)\,\hat{\imath} - \omega\sin(\omega t)\,\hat{\jmath}] \\ &= -r\omega^2[\cos(\omega t)\,\hat{\imath} + \sin(\omega t)\,\hat{\jmath}].\end{aligned}$$

Thus,

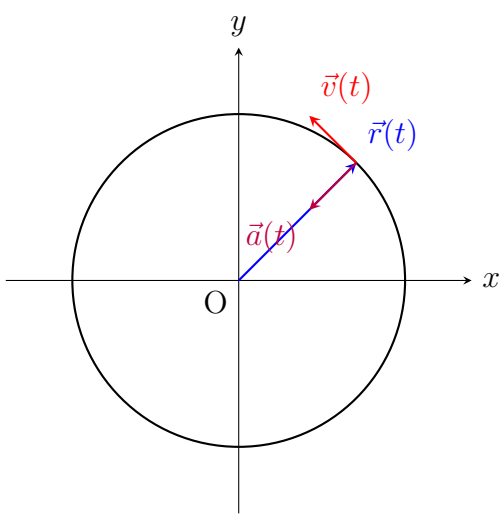
$$\boxed{\vec{a}(t) = -\omega^2\vec{r}(t)}.$$

11.5 Physical Meaning

The acceleration $\vec{a}(t)$ points in the *opposite direction* of $\vec{r}(t)$, i.e. toward the center of the circular path. Hence, it is called the **centripetal acceleration** (from Latin: "center-seeking").

The magnitude of this acceleration is

$$a = |\vec{a}(t)| = \omega^2 r = \frac{v^2}{r}.$$



11.6 Summary of Key Results

$$\vec{r}(t) = r[\cos(\omega t)\hat{\imath} + \sin(\omega t)\hat{\jmath}].$$
$$\vec{v}(t) = r\omega[-\sin(\omega t)\hat{\imath} + \cos(\omega t)\hat{\jmath}].$$
$$\vec{a}(t) = -r\omega^2[\cos(\omega t)\hat{\imath} + \sin(\omega t)\hat{\jmath}] = -\omega^2\vec{r}(t).$$

$$v = r\omega. \qquad a = \frac{v^2}{r} = \omega^2 r.$$

11.7 Conceptual Insight

- The **velocity vector** changes direction continuously, remaining tangent to the circle.
- The **acceleration vector** points toward the center, causing the change in direction of $\vec{v}$, not its magnitude.
- The particle's motion demonstrates that even with constant speed, acceleration can exist if the direction of velocity changes.

In essence: Uniform circular motion is an elegant example of how a purely geometrical change rotation gives rise to acceleration without any change in speed. The particle is always "falling" toward the center, yet never reaches it.

12 Newton's First Law (Law of Inertia)

Statement: *A body continues in its state of rest, or of uniform motion in a straight line, unless acted upon by a net external force.*

This principle expresses a profound truth: constant velocity requires no cause. An object does not need an external agency to keep moving it needs one only to change its state of motion.

12.1 Historical Perspective

12.1.1 Before Newton: The Aristotelian View

Before Newton: The Aristotelian View Before Isaac Newton, the dominant theory of motion was proposed by **Aristotle (384–322 BC)**. According to Aristotle, an object in motion required a **continuous cause** (a constant force) to keep it moving. If the cause was removed, the motion would stop. In his view:

> "Motion ceases when the external cause (force) is removed."

For example, a cart moves because a horse pulls it. Once the horse stops, the cart stops too. Thus, Aristotle believed that rest was the "natural state" of all objects unless acted upon by a force.

12.1.2 Galileo's Challenge to Aristotle

Galileo Galilei (1564–1642) was the first to experimentally challenge Aristotle's ideas. He observed that a ball rolling on a smooth surface slows down mainly due to **friction**. As the surface becomes smoother, the ball rolls farther before stopping.

> In the absence of resistance (such as friction), a moving body would continue to move forever with constant speed.

This insight suggested that motion does *not* require a continuous cause rather, it persists unless an external force (like friction) acts on it.

12.1.3 Newton's Formulation

Isaac Newton (1643–1727) built upon Galileo's findings and expressed this idea formally in his **First Law of Motion**:

Every body continues in its state of rest or uniform motion in a straight lineunless acted upon by a net external force.

This principle introduces the concept of **inertia**the natural tendency of an object to resist any change in its motion.

12.2 Summary Table

12.3 Physical Meaning

Newton's First Law defines **inertial reference frames**coordinate systems where:

$$\sum \vec{F} = 0 \quad \Rightarrow \quad \vec{a} = 0$$

That is, if no net external force acts on a body, its acceleration is zero; it either remains at rest or moves with uniform velocity.

This law provides the foundation for all of Newtonian mechanics.

"Motion does not need a continuous cause; it persists by nature unless something interferes."

Philosopher	Belief About Motion	Example or Interpretation
Aristotle	Motion requires a continuous cause (force). Objects naturally come to rest.	A cart moves only as long as the horse pulls it. Once the pulling stops, motion stops.
Galileo	Motion persists unless opposed by resistance (friction or other forces).	A ball on a frictionless surface would roll forever with constant velocity.
Newton	Motion continues in a straight line with constant velocity unless acted on by a net external force.	A satellite orbits Earth with no engine because there is nearly no resistance in space.

13 Newton's Second Law

Statement: *N*ewton's Second Law links the net force acting on a particle to its motion. In its familiar form:

$$\boxed{\sum \vec{F} = m\,\vec{a}},$$

where $\sum \vec{F}$ is the vector sum of external forces, m is the mass of the particle, and $\vec{a} = d\vec{v}/dt$ is its acceleration.

13.1 Units and the Newton

The SI unit of mass is the kilogram (kg), time is the second (s), and length is the metre (m). From $\sum \vec{F} = m\vec{a}$ we get the derived unit of force:

$$1\ \mathrm{N} = 1\ \mathrm{kg} \cdot \mathrm{m\,s^{-2}},$$

called a *newton* (N). Thus when a force of 1 N acts on a mass of 1 kg, it produces an acceleration of 1 m/s^2.

13.2 How to Test the Law: Measuring Acceleration and Mass

To check the law experimentally we must measure both sides of the equation.

Measuring acceleration. Measure positions at closely spaced times and use finite differences to estimate the velocity and then acceleration. In practice, take three position measurements $x(t_1), x(t_2), x(t_3)$ with $t_3 > t_2 > t_1$, compute the average velocities

$$v_{12} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}, \qquad v_{23} = \frac{x(t_3) - x(t_2)}{t_3 - t_2},$$

and estimate the acceleration as

$$a \approx \frac{v_{23} - v_{12}}{t_3 - t_1}.$$

Making the time intervals smaller and taking the limit recovers the instantaneous acceleration:

$$a(t) = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - 2x(t) + x(t - \Delta t)}{(\Delta t)^2} = \frac{d^2x}{dt^2}.$$

(See the section on limits and instantaneous velocity for more detail.)

Operational (comparative) definition of mass. Mass is not measured directly from the algebraic formula $m = F/a$ unless one already has a calibrated force. Instead, mass is operationally defined by comparison to a reference mass using a repeatable method for applying the same force to different bodies.

Suppose an apparatus (a spring, a compressed launcher, or another device) can apply the same unknown force F to two different objects, one with known reference mass m_{ref} and the other with unknown mass m. If the measured accelerations are a_{ref} and a, then

$$F = m_{\text{ref}} a_{\text{ref}} = ma \quad \Longrightarrow \quad m = m_{\text{ref}} \frac{a_{\text{ref}}}{a}.$$

Thus the unknown mass is found by measuring the ratio of accelerations produced by the same applied force. This is the inertial (dynamical) definition of mass: *mass is the constant of proportionality between a given force and the acceleration it produces.*

Practical implementation (spring / launcher experiment).

1. Anchor one end of a spring to a large laboratory frame (or wall) and stretch or compress it by a fixed amount so the spring exerts a repeatable force F (Hooke's law will relate force to displacement if the spring constant is known, but we do not need to know F's numerical value).
2. Attach a reference mass m_{ref} and release; measure the resulting initial acceleration a_{ref}.
3. Repeat with the test mass m using exactly the same spring displacement; measure a.
4. Compute $m = m_{\text{ref}}(a_{\text{ref}}/a)$.

This method avoids needing a calibrated force sensor: the same (unknown) force acts in both trials and cancels out when taking the ratio.

13.3 Worked numerical example

A reference mass $m_{\text{ref}} = 1$ kg is placed against a launcher that, when released with the same setting, produces an acceleration $a_{\text{ref}} = 2.0$ m/s^2. The test mass shows acceleration

$a = 0.04$ m/s^2 with the identical launcher setting. The test mass is

$$m = 1 \text{ kg} \times \frac{2.0}{0.04} = 50 \text{ kg}.$$

14 Hooke's Law and the Spring Experiment

In an experiment, a spring's elongation x and the acceleration of a known mass are measured. The force exerted by the spring is found from $F = ma$, producing a linear relation:

$$F = -kx$$

where k is the spring constant and the negative sign indicates the restoring nature of the force always opposing displacement.

Important: x represents the change in length from the spring's natural length, not its absolute position.

15 Distinguishing Between $F = ma$ and $F = -kx$

- $F = ma$: Universal law of motion applies to all bodies.
- $F = -kx$: Specific force law describes spring behavior.

Thus, Newton's law gives the framework, while Hooke's law provides the force in a specific situation.

16 Gravitational Force

Near Earth's surface, the gravitational force is:

$$F_g = mg, \qquad a = g.$$

Because $a = F/m = g$, all objects fall with the same acceleration. This reveals that gravitational mass (source of attraction) equals inertial mass (resistance to acceleration)the basis of Einstein's equivalence principle.

17 Applying Newton's Laws: The Power of Prediction

The ultimate goal of mechanics is prediction. Given the state of a system at the present moment (initial conditions), we want to determine its behavior at all future times. To do

this, we combine specific knowledge about forces with Newton's general laws.

17.1 The Structure of the Equation of Motion

Newton's Second Law is often written as $F = ma$, but physically, it represents a bridge between two different types of knowledge:

$$\underbrace{F_{\text{external}}}_{\text{Independent Physics}} = \underbrace{m\frac{d^2x}{dt^2}}_{\text{Universal Response}} \tag{1}$$

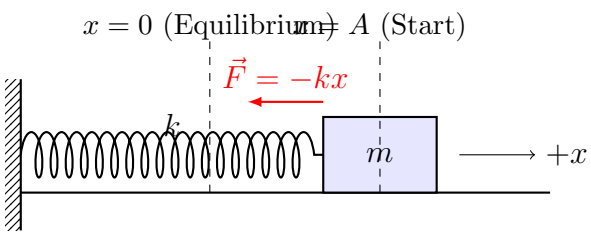
- **The Left-Hand Side (Forces):** This requires specific knowledge about the environment. For example, if we are studying a spring, we must perform experiments to discover Hooke's Law ($F = -kx$). Newton's laws do not tell us what the force is; we must find it separately.
- **The Right-Hand Side (Kinematics):** This represents the object's reaction. Acceleration is the second derivative of position with respect to time.

18 Example: The Spring-Mass System

Consider a mass m on a frictionless table, attached to a spring with force constant k.

18.1 Problem Setup

We pull the mass by a distance A (amplitude) from its equilibrium position ($x = 0$) and release it from rest. We want to know: **What happens next?**



18.2 The Differential Equation

Substituting the spring force into Newton's Second Law:

$$-kx = ma$$

Since $a = \frac{d^2x}{dt^2}$, we get:

$$m\frac{d^2x}{dt^2} = -kx \quad \Rightarrow \quad \frac{d^2x}{dt^2} = -\left(\frac{k}{m}\right)x \tag{2}$$

This is a **Differential Equation**. Unlike algebraic equations where we solve for a number (e.g., $x = 5$), here we must solve for a *function* $x(t)$. We are looking for a function that, when differentiated twice, returns the original function with a negative sign.

18.3 Solving by Guessing (The Ansatz)

In physics, a common method to solve differential equations is to make an educated guess (an ansatz) based on the expected physical behavior.

- **Guess 1: Exponential ($x = e^t$)?**
  Differentiating twice gives e^t. This implies the object flies away faster and faster. This does not match the physical intuition of a spring.
- **Guess 2: Trigonometric ($x = \cos t$)?**
  Differentiating cosine gives negative sine; differentiating again gives negative cosine.

  $$\frac{d}{dt}(\cos t) = -\sin t, \quad \frac{d^2}{dt^2}(\cos t) = -\cos t$$

  This matches the form of our equation ($a \propto -x$). Physically, cosine represents **oscillation**, which is exactly what a spring does.

To fit the initial condition (that we pulled it to distance A), we scale the solution:

$$x(t) = A\cos(\omega t) \tag{3}$$

where ω relates to k and m.

19 Refining the Model: The Role of Friction

Our mathematical solution $x(t) = A\cos(\omega t)$ suggests the mass will oscillate forever with constant amplitude. However, experience tells us the mass eventually stops.

Does this mean Newton's Laws are wrong? **No.**

It means our **Physical Model** (the Left-Hand Side) was incomplete. We assumed the spring force was the *only* force. To match reality, we must add a friction force, updating our equation:

$$\underbrace{-kx}_{\text{Spring}} + \underbrace{F_{\text{friction}}}_{\text{Missing Force}} = ma$$

This illustrates the iterative nature of physics: we compare the mathematical prediction to the experimental reality to refine our understanding of the forces at play.

20 Friction

In ideal scenarios, we ignore resistance. However, in the real world, objects stop moving due to **Friction**. Friction is a resistive force acting parallel to the surface of contact, opposing relative motion.

20.1 Static Friction (f_s)

Static friction acts when there is **no relative motion** between surfaces. It is a variable force.

- If you push a heavy desk with 1 N of force and it doesn't move, the static friction is exactly 1 N (canceling your push).
- If you push with 10 N and it still doesn't move, static friction has increased to 10 N.

It behaves like a "smart" force until it reaches a breaking point:

$$f_s \leq \mu_s N \tag{4}$$

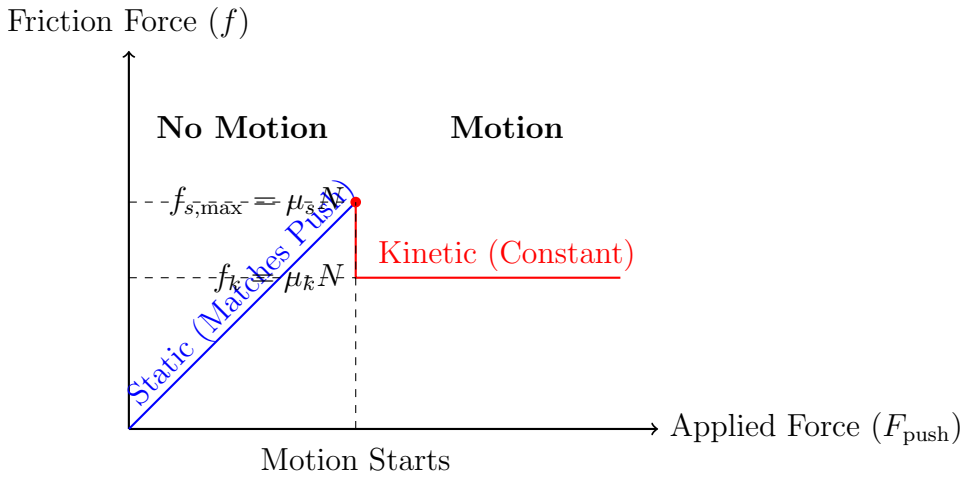
where μ_s is the coefficient of static friction and N is the normal force.

20.2 Kinetic Friction (f_k)

Once the object starts sliding, the force becomes constant and is usually weaker than the maximum static friction:

$$f_k = \mu_k N \tag{5}$$

where μ_k is the coefficient of kinetic friction ($\mu_k < \mu_s$).



The Friction Graph: Notice how static friction rises linearly to match the applied force, then drops to a lower constant value once motion begins.

21 The Inclined Plane

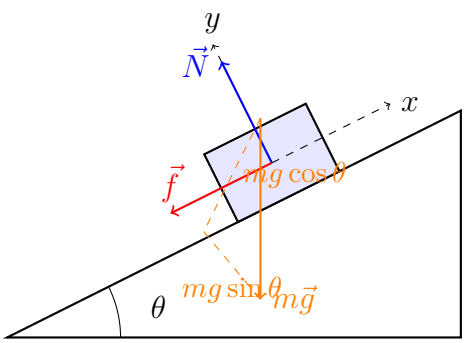
The inclined plane is a classic problem that teaches the importance of choosing the right coordinate system.

21.1 Resolving Forces

Instead of using horizontal and vertical axes, we align our axes with the motion:

- **x-axis:** Parallel to the incline (downhill is positive).
- **y-axis:** Perpendicular to the incline.

Since gravity acts vertically downwards, we must resolve it into components relative to these new axes.



21.2 Equations of Motion

From the diagram, we write Newton's Second Law for each axis:

1. Perpendicular Direction (y): The block does not fly off or sink into the ramp, so $a_y = 0$.

$$\sum F_y = N - mg\cos\theta = 0 \implies N = mg\cos\theta$$

2. Parallel Direction (x): Assuming the block accelerates down the slope with friction:

$$\sum F_x = mg\sin\theta - f_k = ma_x$$

Substituting friction $f_k = \mu_k N = \mu_k(mg\cos\theta)$:

$$mg\sin\theta - \mu_k mg\cos\theta = ma$$

The mass m cancels out, yielding:

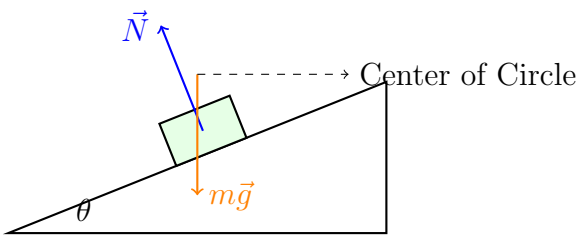
$$a = g(\sin\theta - \mu_k\cos\theta) \tag{6}$$

22 Dynamics of Circular Motion

When an object moves in a circle of radius R at speed v, it experiences a centripetal acceleration $a_c = v^2/R$ directed toward the center. Newton's Law states that a net force must provide this acceleration.

22.1 The Banked Turn (No Friction)

Cars can turn on curved roads without friction if the road is "banked" (tilted) at an angle θ. The Normal force provides the necessary centripetal force.



We resolve forces into horizontal (radial) and vertical directions:

1. **Vertical:** The car doesn't accelerate up or down.

$$N\cos\theta = mg$$

2. **Horizontal:** The component of N pushes toward the center.

$$N \sin\theta = \frac{mv^2}{R}$$

Dividing the second equation by the first:

$$\frac{N \sin\theta}{N \cos\theta} = \frac{mv^2/R}{mg} \implies \tan\theta = \frac{v^2}{Rg}$$

This formula tells civil engineers exactly how much to tilt a road for a given speed limit v and radius R.

22.2 Loop-the-Loop

Consider a roller coaster doing a vertical loop. At the very top, both gravity and the track's normal force point **down**.

$$\sum F_{\text{down}} = N + mg = \frac{mv^2}{R}$$

To complete the loop without falling, the car must maintain contact with the track, meaning $N \geq 0$. Setting $N = 0$ gives the minimum speed required:

$$mg = \frac{mv^2}{R} \implies v_{\text{min}} = \sqrt{Rg}$$

If $v < \sqrt{Rg}$, the car falls off the track at the top.

23 Banked Turn: Physical Meaning

When a car moves on a flat curved road, friction provides the centripetal force needed to keep it in circular motion. However, on a **banked** (tilted) road, part of the normal force contributes to the centripetal force.

Intuitive Explanation

Tilting the road causes the normal force to tilt as well. The normal force N then has two components:

$$N \cos\theta \quad \text{(vertical)}, \qquad N \sin\theta \quad \text{(horizontal)}$$

The horizontal component points toward the center of the circular path and provides the centripetal force required for the turn:

$$N \sin\theta = \frac{mv^2}{R}$$

The vertical component balances the weight:

$$N \cos\theta = mg$$

Dividing the two equations gives the design condition:

$$\tan\theta = \frac{v^2}{Rg}$$

This means that at the *correct* banking angle θ, the car can make the turn **without relying on friction**. The tilt itself provides the needed inward force.

Physical Meaning

The equation tells :

- Larger speeds require larger banking angles.
- Sharper turns (small R) require larger banking angles.
- A perfectly banked curve needs zero friction.

24 Loop-the-Loop: Physical Meaning

A car or roller coaster moving through a vertical loop experiences different force conditions at various points in the loop. The most critical point is the **top of the loop**.

Intuitive Explanation

At the top of the loop, both gravity and the normal force point downward, toward the center of the circular path. For the car to remain in contact with the track, the normal force must be non-negative:

$$N \geq 0$$

At the *minimum safe speed*, the car is just about to lose contact, so:

$$N = 0$$

Then gravity must alone supply the centripetal force:

$$mg = \frac{mv^2}{R}$$

This gives the minimum required speed at the top:

$$v_{\text{min}} = \sqrt{Rg}$$

Physical Meaning

- If the car's speed is less than $\sqrt{Rg}$ at the top, it will lose contact and fall.
- At exactly $\sqrt{Rg}$, gravity alone is sufficient to produce the needed centripetal acceleration.
- Real roller coasters start at a higher height to ensure the speed is well above this minimum.

Summary

- **Banked Turn:** Tilting the road redirects part of the normal force to act as centripetal force.
- **Loop-the-Loop:** At the top of the loop, the minimum speed occurs when gravity alone provides the centripetal force.

25 The Loop-the-Loop Paradox

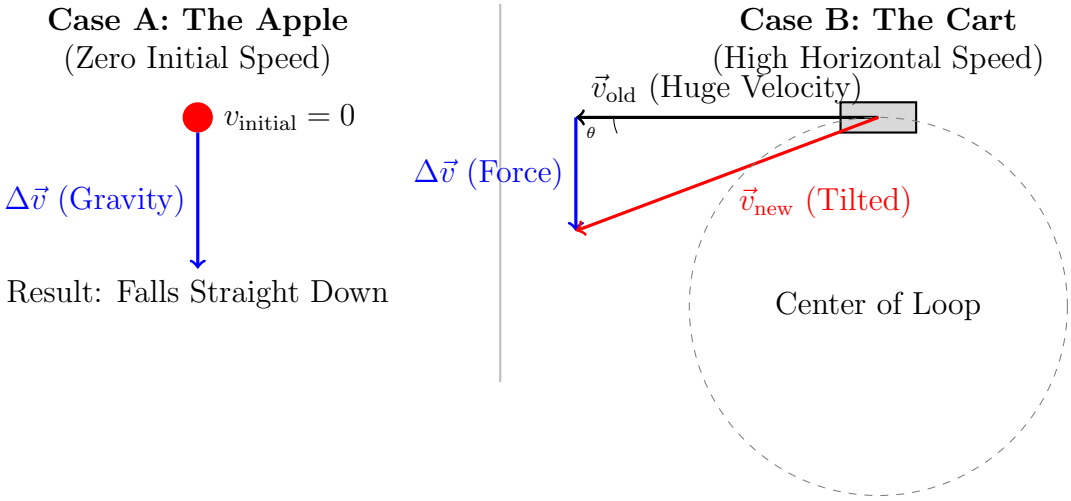
25.1 The Bullet vs. The Apple

Imagine holding an apple and a gun at the same height.

- **The Apple:** If you drop it, it has zero initial speed. Gravity pulls it straight down. It falls vertically.
- **The Bullet:** If you fire a bullet horizontally, gravity still pulls it down exactly as hard as the apple. However, the bullet is moving so fast sideways that by the time it falls 1 inch, it has traveled 1 mile horizontally.

The Loop-the-Loop cart is like the bullet. It is *falling* towards the center, but it is moving sideways so fast that the floor (the track) curves away beneath it.

25.2 Visualizing the Vector Addition



Comparison of falling from rest vs. falling with high tangential velocity. In Case B, the velocity vector changes direction (tilts) rather than pointing straight down.

25.3 Physical Meaning

1. "Falling Around" the Center
When the cart is at the top, it is technically accelerating downward (falling). However, "falling" does not mean "getting closer to the ground."

- Because the velocity is perpendicular to the force, the force acts only to **turn** the velocity vector, not to speed it up or slow it down.
- The cart falls towards the center, but the circular track curves away at the exact same rate.

2. The Math of "Staying Up"
For the cart to track the circle, the downward forces (Gravity + Normal Force) must provide the exact centripetal acceleration required to turn the velocity vector:

$$mg + N = \frac{mv^2}{R}$$

If v is too slow, the "turn" required is not sharp enough, gravity pulls it down faster than the circle curves, and the cart falls off (N becomes zero).

26 Introduction to Energy

We are now transitioning from the study of Forces (Dynamics) to the study of Energy. This shift is not just mathematical; it represents a deeper way of understanding the universe.

26.1 Why a New Concept?

Newton's Laws ($\vec{F} = m\vec{a}$) allow us to predict the motion of objects, but they rely on specific details that are not always available:

- Newton's laws deal with **Vectors** (direction matters).
- They assume particles have definite **trajectories** (precise position and velocity at all times).

However, as physics advanced into the sub-atomic realm (**Quantum Mechanics**), we discovered that particles like electrons do not have definite trajectories. The concepts of precise position and acceleration break down. Newton's laws are "surrendered" in this domain.

26.2 The Robustness of Energy

While the concept of "force" and "trajectory" fails in quantum mechanics, the concept of **Energy** survives. Energy is a **Scalar** quantity (a number, not a vector). It turns out to be a more fundamental property of nature than force.

> "The notion of energy turns out to be very robust and survives all the quantum revolution."

26.3 Historical Context: The Neutrino

The physicist's faith in the Conservation of Energy is best illustrated by the discovery of the neutrino.

1. In the early 20th century, physicists observed nuclear beta decay and found that the energy of the products was less than the energy of the starting material. Energy seemed to vanish.
2. **Niels Bohr** suggested that perhaps the Law of Conservation of Energy was only statistically true and could be violated in individual nuclear events.

3. **Wolfgang Pauli** disagreed. He postulated that the "missing" energy was actually being carried away by a tiny, electrically neutral, invisible particle that detectors couldn't see.

Pauli was right. Years later, this particlethe **Neutrino**was detected. Physicists preferred to invent a new, invisible particle rather than abandon the Law of Conservation of Energy.

26.4 Philosophy of Derivation

We will not simply define Kinetic Energy as $K = \frac{1}{2}mv^2$. Instead, we will follow the logical path that leads us to this quantity. We will see that energy arises naturally when we integrate Newton's laws over a distance.

27 The Work-Energy Theorem

Newton's Second Law ($F = ma$) describes how forces cause acceleration. However, by integrating this law over distance, we derive a powerful new tool relating force to speed.

27.1 Constant Force in 1D

Consider a constant force F pushing a mass m over a distance d. From kinematics ($v^2 = v_0^2 + 2ad$) and Newton's law ($a = F/m$):

$$v_f^2 = v_i^2 + 2\left(\frac{F}{m}\right)d$$

Rearranging terms to group mass and velocity together:

$$\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = F \cdot d$$

This equation introduces two new physical quantities:

- **Kinetic Energy (K):** defined as $K = \frac{1}{2}mv^2$.
- **Work (W):** defined as $W = F \cdot d$ (for constant force).

The **Work-Energy Theorem** states that the net work done on an object equals its change in kinetic energy:

$$W_{\text{net}} = \Delta K = K_f - K_i \tag{7}$$

27.2 Variable Force and Calculus

In reality, forces often vary with position (e.g., a spring $F(x) = -kx$). In such cases, we cannot simply multiply $F \times d$. Instead, we must sum the work done over infinitesimally small displacements dx:

$$dW = F(x)dx$$

The total work is the integral (the area under the force-position graph):

$$W = \int_{x_1}^{x_2} F(x)dx \tag{8}$$

28 Conservative Forces and Potential Energy

28.1 Defining Potential Energy (U)

We can simplify the integral by introducing a function $U(x)$, called **Potential Energy**, defined such that the force is the negative derivative of the potential:

$$F(x) = -\frac{dU}{dx} \tag{9}$$

Conversely, the change in potential energy is the negative of the work done by the force:

$$\Delta U = U(x_2) - U(x_1) = -\int_{x_1}^{x_2} F(x)dx = -W$$

28.2 Common Examples

1. **Gravity (near Earth):** Force $F = -mg$ (downward).

   $$U(y) = mgy$$

   Checking: $-\frac{d}{dy}(mgy) = -mg$. (Correct).

2. **Spring Force:** Force $F = -kx$ (restoring force).

   $$U(x) = \frac{1}{2}kx^2$$

   Checking: $-\frac{d}{dx}(\frac{1}{2}kx^2) = -kx$. (Correct).

29 Conservation of Mechanical Energy

Substituting the potential energy definition into the Work-Energy Theorem:

$$W = \Delta K$$

$$-(U_2 - U_1) = K_2 - K_1$$

Rearranging terms to group initial and final states:

$$K_1 + U_1 = K_2 + U_2 \tag{10}$$

This sum is called the **Total Mechanical Energy (E)**:

$$E = K + U = \text{constant}$$

Physical Meaning: Even though the object's speed (v) and position (x) are constantly changing, the sum of Kinetic and Potential energy remains constant (conserved) throughout the motion, provided no non-conservative forces (like friction) are doing work.

This allows us to solve problems (like finding impact speed) without needing to know the time or the detailed acceleration history.

30 Applying Conservation of Energy

The true power of the Conservation of Energy lies in its ability to solve complex problems without integration.

30.1 Example: The Spring System

Consider a mass on a spring pulled to amplitude A and released.

- **Using Newton's Laws:** The force $F = -kx$ changes continuously, meaning acceleration changes continuously. Finding $v(t)$ requires solving a differential equation.
- **Using Energy:** We know the total energy at the start is entirely potential: $E = \frac{1}{2}kA^2$. At any position x, the energy is shared between kinetic and potential:

  $$\frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

  We can instantly solve for velocity v:

  $$v = \sqrt{\frac{k}{m}(A^2 - x^2)}$$

30.2 Multiple Forces

If multiple conservative forces act on a body (e.g., a vertical spring affected by gravity), we simply sum the potential energies:

$$E_{\text{total}} = K + U_{\text{gravity}} + U_{\text{spring}}$$

$$E = \frac{1}{2}mv^2 + mgy + \frac{1}{2}ky^2 = \text{Constant}$$

This sum remains invariant regardless of how the object moves.

31 The Problem with Friction

Friction is a non-conservative force (often called a "bad" force in this context). Unlike gravity or springs, we cannot define a potential energy function $U(x)$ for friction.

31.1 The Fundamental Forces are Conservative

It is important to note that friction is a macroscopic approximation. At the fundamental level (atoms interacting), the forces are electromagnetic and *are* conservative. Friction appears non-conservative only because we ignore the complex microscopic motion of trillions of atoms (heat) and look only at the bulk object.

31.2 Why No Potential for Friction?

Potential energy relies on the force being a function of position only ($F(x)$).

- Gravity acts down regardless of whether you move up or down.
- Friction, however, opposes **velocity**. If you move right, it points left. If you move left, it points right.
- Since the force depends on direction of motion, it cannot be written as a simple derivative of a position function $U(x)$.

31.3 Modified Work-Energy Theorem

Instead of conserving total mechanical energy, friction dissipates it (turns it into heat). We account for this by calculating the work done by friction explicitly:

$$E_{\text{final}} - E_{\text{initial}} = W_{\text{friction}}$$

Since friction always opposes motion, W_{friction} is negative:

$$W_{\text{friction}} = -f_k d$$

Thus:

$$E_{\text{final}} = E_{\text{initial}} - f_k d \tag{11}$$

32 Mathematical Interlude: Approximations

In physics, we often deal with small changes (Δx) where functions can be approximated as straight lines. This is based on the Taylor Series expansion.

32.1 The Linear Approximation

For any function $f(x)$, the value at a nearby point $x + \Delta x$ is approximately:

$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x$$

This states that the new value is the old value plus the slope times the step size.

32.2 The Binomial Approximation

A specific and very useful case is the power function $(1 + x)^n$. If $x \ll 1$, then:

$$(1 + x)^n \approx 1 + nx \tag{12}$$

Example: Relativistic Mass
In relativity, mass increases with speed v:

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}} = m_0 \left(1 - \frac{v^2}{c^2}\right)^{-1/2}$$

If $v \ll c$ (speed is much less than light), let $x = -v^2/c^2$ and $n = -1/2$. Using the approximation:

$$m \approx m_0 \left(1 + \left(-\frac{1}{2}\right)\left(-\frac{v^2}{c^2}\right)\right) = m_0 + \frac{1}{2} m_0 \frac{v^2}{c^2}$$

This formula bridges the gap between Newtonian physics and Einstein's relativity for slow-moving objects.

33 The Logic of Approximations

we often ask: *"Why do we approximate? Why not just use the exact formula?"* In physics, approximations are not just about avoiding difficult math; they provide physical insight.

33.1 Why We Need Them

1. **Simplification:** Many physical systems (like a pendulum) have complex non-linear equations. However, for small movements (small angles), these equations become linear and solvable.

2. **Bridging Theories:** Approximations show how advanced theories connect to basic ones. For example, using the binomial approximation on Einstein's relativistic energy equation reveals that it contains Newton's kinetic energy ($\frac{1}{2}mv^2$) hidden inside it.

33.2 Linear vs. Binomial: What is the difference?

1. Linear Approximation (The General Rule)
This is the fundamental concept from calculus (Taylor Series). It states that *any* smooth function $f(x)$ looks like a straight line if you zoom in close enough.

$$f(x + \Delta x) \approx f(x) + \left(\frac{df}{dx}\right) \Delta x$$

This works for sine, cosine, exponentials, etc.

2. Binomial Approximation (The Specific Shortcut)
This is a **special case** of the linear approximation applied specifically to power functions of the form $(1 + x)^n$.

- Instead of taking the derivative every time, we use the shortcut: **"Bring the power down."**
- **Formula:** $(1 + x)^n \approx 1 + nx$ (valid when $x \ll 1$).

33.3 Example of the Connection

Let's derive the Binomial shortcut using the Linear method. Let $f(u) = u^n$. We want to find the value near $u = 1$.

$$f(1 + x) \approx f(1) + f'(1) \cdot x$$

- $f(1) = 1^n = 1$.
- The derivative is $f'(u) = nu^{n-1}$. Evaluated at $u = 1$, this is $n(1)^{n-1} = n$.

Substituting these back gives the Binomial result:

$$(1 + x)^n \approx 1 + n \cdot x$$

This confirms that the Binomial approximation is just the Linear approximation applied to powers.

34 Energy in Higher Dimensions

Moving from 1D to 2D requires us to generalize our mathematical tools. We now deal with functions of multiple variables $f(x, y)$ and vectors with multiple components.

34.1 Calculus of Many Variables

A function $f(x, y)$ represents a surface above the xy-plane (like terrain height). To measure how this function changes, we introduce the **Partial Derivative**.

- **Partial with respect to x ($\frac{\partial f}{\partial x}$):** The rate of change of f as we move along x, holding y constant.
- **Partial with respect to y ($\frac{\partial f}{\partial y}$):** The rate of change of f as we move along y, holding x constant.

The total change in the function for a small displacement is:

$$\Delta f \approx \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y \tag{13}$$

Equality of Mixed Partials:
For smooth functions, the order of differentiation does not matter:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

This reflects the geometric fact that moving Δx then Δy brings you to the same point as moving Δy then Δx.

34.2 Deep Dive: Equality of Mixed Partials

often wonder why the order of differentiation doesn't matter.

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

1. A Concrete Example
Let's test this with a function where x and y interact: $f(x, y) = x^2y^3$.

Case A: Differentiate with respect to x, then y.

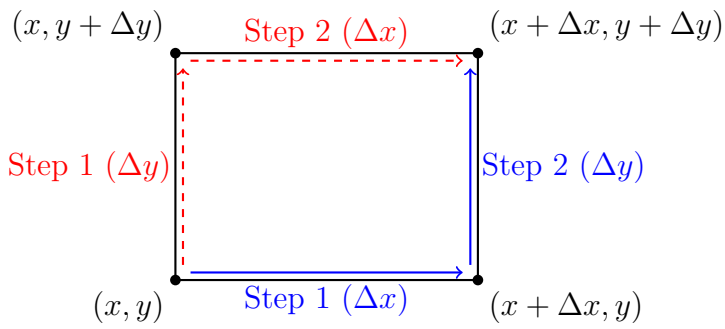
$$\begin{aligned} \text{Step 1 } (x)\text{:} \quad & \frac{\partial}{\partial x}(x^2y^3) = 2xy^3 \\ \text{Step 2 } (y)\text{:} \quad & \frac{\partial}{\partial y}(2xy^3) = 2x(3y^2) = \mathbf{6xy^2} \end{aligned}$$

Case B: Differentiate with respect to y, then x.

$$\begin{aligned} \text{Step 1 } (y)\text{:} \quad & \frac{\partial}{\partial y}(x^2y^3) = x^2(3y^2) = 3x^2y^2 \\ \text{Step 2 } (x)\text{:} \quad & \frac{\partial}{\partial x}(3x^2y^2) = 3(2x)y^2 = \mathbf{6xy^2} \end{aligned}$$

Both methods yield the exact same result.

2. The Geometric Reason (The Rectangle)
using path independence. Imagine trying to measure the change in height Δf between point A and point C.



- **Blue Path:** Change x first, then change y.
- **Red Path:** Change y first, then change x.

Since $f(x, y)$ represents a physical surface (like a mountain), the height at the final corner $(x + \Delta x, y + \Delta y)$ is a fixed physical reality. It cannot depend on which path you took to get there. Therefore, the accumulation of changes (the derivatives) must be equal regardless of the order.

35 Work and the Dot Product

35.1 Deriving Work in 2D

We want to preserve the Work-Energy Theorem: $W = \Delta K$. Kinetic energy in 2D is $K = \frac{1}{2}m(v_x^2 + v_y^2)$. Differentiating with respect to time gives the Power (P):

$$P = \frac{dK}{dt} = mv_x a_x + mv_y a_y = F_x v_x + F_y v_y$$

Multiplying by dt gives the work done:

$$dW = F_x dx + F_y dy$$

35.2 The Dot Product

The structure $A_x B_x + A_y B_y$ appears frequently in physics. We define it as the **Scalar Product** or **Dot Product**:

$$\vec{A} \cdot \vec{B} \equiv A_x B_x + A_y B_y + A_z B_z \tag{14}$$

Geometric Interpretation: Using the Law of Cosines, it can be proven that:

$$\vec{A} \cdot \vec{B} = |A||B| \cos\theta \tag{15}$$

where θ is the angle between the vectors.

Physical Significance: The dot product is an **Invariant Scalar**. If you rotate your coordinate axes, the individual components A_x, A_y change, but the value $\vec{A} \cdot \vec{B}$ remains constant. This is crucial because physical work shouldn't depend on which way we hold our ruler.

35.3 The Line Integral

Work is defined as the integral of the force along a specific path C from point A to point B:

$$W_{A \to B} = \int_A^B \vec{F} \cdot d\vec{r} \tag{16}$$

This is a **Line Integral**. We chop the path into tiny vectors $d\vec{r}$, calculate the work $dW = \vec{F} \cdot d\vec{r}$ for each segment (taking into account that the force might not be parallel to the path), and sum them up.

36 Deep Dive: The Line Integral

The concept of work $W = F \cdot d$ works only for straight lines and constant forces. How do we calculate work when the object moves along a curved path C through a changing force field?

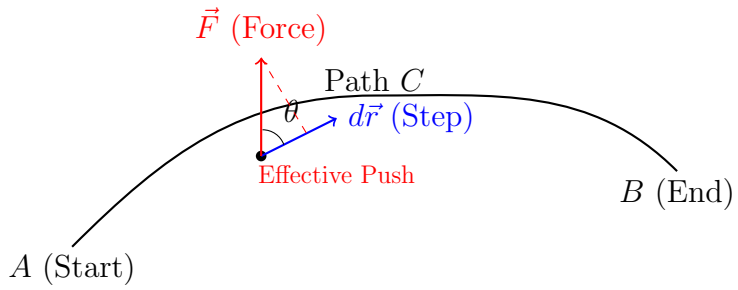
36.1 The "Chop, Dot, and Sum" Method

Imagine a particle moving along a curved path from point A to point B.

1. **Chop:** We divide the curved path into infinite tiny steps. Each step is a vector $d\vec{r}$ (pointing tangent to the path).
2. **Dot:** At each step, the force $\vec{F}$ might point in a different direction than the path. We use the **Dot Product** to find the component of force *parallel* to the motion.

$$dW = \vec{F} \cdot d\vec{r} = |\vec{F}||d\vec{r}| \cos\theta$$

3. **Sum:** We add up (integrate) the work contributions from every tiny step along the path.



figureVisualizing the Line Integral. At every point, we only care about the part of the Force that points along the direction of the step ($d\vec{r}$).

36.2 The Formula

The total work is written as:

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_A^B (F_x dx + F_y dy) \tag{17}$$

This equation says: "Walk along curve C. At every step, multiply the Force by the step distance, accounting for the angle θ. Then sum it all up."

37 Differentials vs. Integrals: A Physical Perspective

To understand why Energy Conservation works (or fails), we must distinguish between local changes and global accumulations.

37.1 The Differential: The "Local" Reality

Differential quantities (like $d\vec{r}$, dt, dW) represent physics occurring at a single instant in time or a single point in space.

- **Definition:** What is happening *right now*?
- **Example:** Power ($P = \vec{F} \cdot \vec{v}$) tells us the rate of energy transfer at this exact second. It does not care about the past.

37.2 The Integral: The "Global" History

Integral quantities ($\int$) represent the accumulation of effects over an extended period or distance.

- **Definition:** What happened over the *entire journey*?
- **Example:** Work ($W = \int \vec{F} \cdot d\vec{r}$) tells us the total energy change after the trip is finished.

38 Conservative Forces: When does Potential Energy Exist?

In 1D, every force (that depends only on position) has a potential energy. In 2D and 3D, this is not true.

38.1 The Problem of Path Dependence

If we want to define potential energy $U(\vec{r})$, the work done in moving from point A to point B must depend **only** on the endpoints A and B, and not on the path taken.

$$\int_{A,\text{Path 1}}^{B} \vec{F} \cdot d\vec{r} \stackrel{?}{=} \int_{A,\text{Path 2}}^{B} \vec{F} \cdot d\vec{r}$$

For a general random force $\vec{F}(x, y)$, this equality does not hold. If the work depends on the path, we cannot assign a single energy value to a location. Such forces are called **Non-Conservative**.

38.2 The Test for Conservative Forces

A force is conservative if and only if it can be written as the gradient of a scalar potential function U:

$$F_x = -\frac{\partial U}{\partial x}, \quad F_y = -\frac{\partial U}{\partial y}$$

Using the equality of mixed partial derivatives ($\frac{\partial^2 U}{\partial x \partial y} = \frac{\partial^2 U}{\partial y \partial x}$), we can derive a simple test for any force $\vec{F} = (F_x, F_y)$.

The Condition:

$$\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x} \tag{18}$$

If a force satisfies this condition, a potential energy function $U(x, y)$ exists, and mechanical energy is conserved. If not, the force is non-conservative.

38.3 Application: The Roller Coaster

Consider a roller coaster moving on a curved track under gravity.

- **Gravity:** We know $\vec{F}_g = (0, -mg)$. Testing the condition: $\frac{\partial(0)}{\partial y} = 0$ and $\frac{\partial(-mg)}{\partial x} = 0$. It is conservative.
- **Normal Force:** The track exerts a normal force $\vec{N}$ constraint to the path. Does it do work?
$$dW_N = \vec{N} \cdot d\vec{r}$$
Since the normal force is always perpendicular to the displacement ($\vec{N} \perp d\vec{r}$), the dot product is zero.

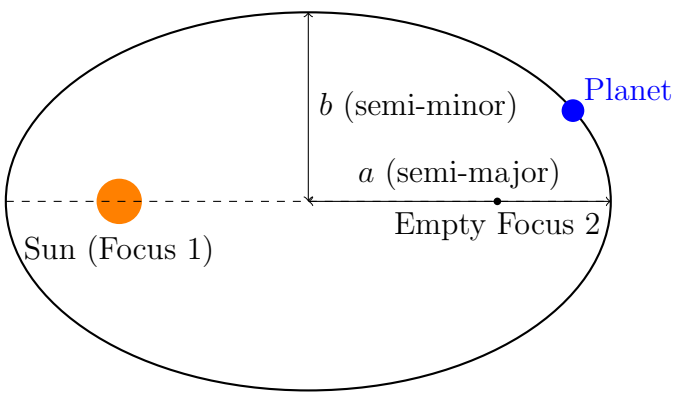
Conclusion: The track forces do **zero work**. Therefore, the total mechanical energy is conserved:
$$E = \frac{1}{2}mv^2 + mgy = \text{Constant}$$
This allows us to determine the speed of the cart at any height y simply by knowing its initial energy.

39 Kepler's Laws of Planetary Motion

Johannes Kepler, after analyzing 40 years of data collected by Tycho Brahe, deduced three fundamental laws that govern the motion of planets. This was a shift from "Terrestrial Dynamics" to "Celestial Mechanics."

39.1 First Law: The Law of Ellipses

Statement: All planets move in elliptical orbits, with the Sun situated at one of the focal points (foci).

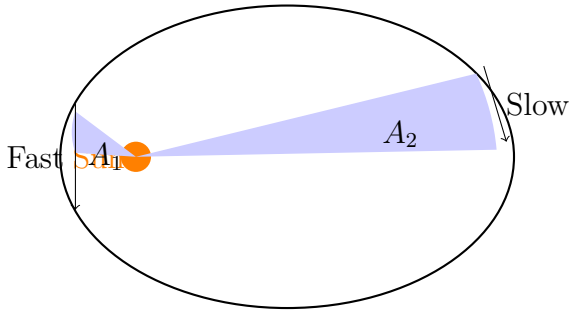


An elliptical orbit. Note that the Sun is off-center.

39.2 Second Law: The Law of Equal Areas

Statement: A line joining a planet and the Sun sweeps out equal areas during equal intervals of time.

$$\frac{dA}{dt} = \text{Constant}$$



If $\Delta t_1 = \Delta t_2$, then Area $A_1 = A_2$

Physical Implication: Planets move faster when they are closer to the Sun (Perihelion) and slower when they are farther away (Aphelion).

39.3 Third Law: The Law of Periods

Statement: The square of the orbital period (T) is directly proportional to the cube of the semi-major axis (a) of its orbit.

$$T^2 \propto a^3 \quad \text{or} \quad \frac{T^2}{a^3} = \text{Constant} \tag{19}$$

This law connects the motions of all planets in the solar system. It implies that distant planets have significantly longer years, not just because they have a longer path to travel, but because they travel slower.

39.4 The Transition to Newton

Kepler provided the **Kinematics** of the solar system (describing *how* planets move). He did not know *why*. Isaac Newton later used these laws to derive the Law of Universal Gravitation, showing that these specific motions are caused by a force that drops off as $1/r^2$.

40 Universal Gravitation

Newton unified terrestrial physics (apples) and celestial physics (moons) by realizing they are governed by the same force.

40.1 The Inverse Square Law

Newton compared the acceleration of an apple near Earth's surface ($a_A = 9.8$ m/s^2) to the centripetal acceleration of the Moon ($a_M = v^2/r$).

- The Moon is 60 times further from the Earth's center than the apple ($r_M = 60R_E$).
- He found that $a_A/a_M \approx 3600 = 60^2$.

This precise correlation implied that the force of gravity weakens with the square of the distance:

$$F_g = G\frac{Mm}{r^2} \tag{20}$$

where G is the universal gravitational constant.

40.2 Gravitational Potential Energy

For celestial objects, the force is not constant, so $U \neq mgh$. Since the force is conservative, we define potential energy by integrating the force from infinity (where force is zero) to a distance r.

$$U(r) = -\frac{GmM}{r} \tag{21}$$

Why Negative? We define $U = 0$ at $r = \infty$. As a mass falls toward a planet, it loses potential energy (and gains kinetic). Since it started at 0 and went down, the potential energy must be negative.

40.3 Reconciling mgh and $-GmM/r$

Near Earth's surface, we are at distance $r = R_E + h$, where $h \ll R_E$. Using the exact formula:

$$U(h) = -\frac{GmM}{R_E + h} = -\frac{GmM}{R_E(1 + h/R_E)}$$

Using the Binomial Approximation $(1 + x)^{-1} \approx 1 - x$:

$$U(h) \approx -\frac{GmM}{R_E}\left(1 - \frac{h}{R_E}\right) = \underbrace{-\frac{GmM}{R_E}}_{\text{Constant } U_0} + \underbrace{\left(\frac{GmM}{R_E^2}\right)}_{\text{equals } mg} h$$

Thus, $U(h) = U_0 + mgh$. Since we only measure *changes* in potential energy, the constant U_0 drops out, and we recover the familiar mgh.

40.4 Escape Velocity

To escape a planet's gravity completely, an object must have enough kinetic energy to reach infinity with at least zero velocity.

$$E_{\text{total}} = K + U = \frac{1}{2}mv^2 - \frac{GmM}{R} \geq 0$$

Solving for the minimum velocity (v_{esc}):

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} \tag{22}$$

For Earth, this is approx 11.2 km/s. If $v < v_{\text{esc}}$, the object is gravitationally bound (it orbits or falls back). If $v \geq v_{\text{esc}}$, it escapes forever.

40.5 Application: Dark Matter

By observing the orbital speeds v of stars at the edge of galaxies and using Newton's orbit formula $v^2 = GM/r$, we can calculate the mass M of the galaxy.

- **Observation:** Stars move much faster than expected based on the visible matter (stars/gas).
- **Conclusion:** There must be a massive amount of invisible matter providing extra gravity. This is the evidence for **Dark Matter**.

41 Dynamics of Multi-Body Systems

When dealing with systems of multiple particles, tracking every individual force becomes impossible. By defining the **Center of Mass**, we can simplify the entire system into a single point that obeys Newton's Laws.

41.1 Derivation for Two Bodies

Consider two masses m_1 and m_2 at positions x_1 and x_2. The forces acting on them are:

- **External Forces (F_{ext}):** Forces from outside the system (e.g., gravity).
- **Internal Forces (F_{int}):** Forces the particles exert on each other (e.g., a spring or collision).

Applying Newton's Second Law to each particle:

$$m_1\ddot{x}_1 = F_{1,ext} + F_{12} \tag{23}$$
$$m_2\ddot{x}_2 = F_{2,ext} + F_{21} \tag{24}$$

where $\ddot{x}$ denotes acceleration (d^2x/dt^2) and F_{12} is the force on 1 due to 2.

Summing the Equations: Adding the two equations together:

$$m_1\ddot{x}_1 + m_2\ddot{x}_2 = (F_{1,ext} + F_{2,ext}) + \underbrace{(F_{12} + F_{21})}_{0}$$

According to Newton's Third Law, action equals reaction ($F_{12} = -F_{21}$), so the internal forces cancel out perfectly.

41.2 The Center of Mass Equation

We define the Center of Mass coordinate X_{CM} as:

$$X_{CM} = \frac{m_1x_1 + m_2x_2}{M_{total}} \tag{25}$$

Differentiating twice gives the acceleration of the Center of Mass:

$$M_{total}\ddot{X}_{CM} = m_1\ddot{x}_1 + m_2\ddot{x}_2$$

Substituting this back into our summed force equation gives the fundamental result:

$$F_{ext}^{net} = M_{total}\ddot{X}_{CM} \tag{26}$$

41.3 Physical Interpretation

The Center of Mass of a system moves exactly like a single point particle of mass M acted upon by the net external force.

- **Example 1 (Projectile):** If you throw a spinning wrench, the handle and head wobble chaotically, but the Center of Mass traces a perfect parabola.
- **Example 2 (Explosion):** If a projectile explodes in mid-air, the internal forces send fragments flying in all directions. However, since internal forces cancel out, the Center of Mass of all the fragments continues along the original trajectory.

42 Calculating Center of Mass for Continuous Bodies

For solid objects, we replace the sum $\sum m_ix_i$ with an integral $\int x\,dm$.

42.1 The Uniform Rod

Consider a rod of length L and total mass M.

- Mass per unit length (Linear Density): $\lambda = M/L$.
- We choose a small slice of width dx. Its mass is $dm = \lambda dx$.

The CM position is:

$$X_{CM} = \frac{1}{M}\int_0^L x\,dm = \frac{1}{M}\int_0^L x\left(\frac{M}{L}\right)dx = \frac{1}{L}\left[\frac{x^2}{2}\right]_0^L = \frac{L}{2}$$

(As expected by symmetry).

42.2 The Triangle

Consider a triangle of base b and height h. We want to find the CM distance from the top vertex. We slice it into horizontal strips of width dx at a distance x from the top. Using similar triangles, the width of the strip w at depth x is:

$$\frac{w}{b} = \frac{x}{h} \implies w = \frac{b}{h}x$$

The mass of the strip is proportional to its area: $dm \propto w \cdot dx$. Integrating $x \cdot dm$ yields the result:

$$X_{CM} = \frac{2}{3}h \quad \text{(from the vertex)} \tag{27}$$

43 Conservation of Momentum

Newton's Second Law for a system is $\vec{F}_{ext} = \frac{d\vec{P}_{total}}{dt}$.

- **The Law:** If the net external force is zero ($\vec{F}_{ext} = 0$), the total momentum is conserved (constant).

$$\vec{P}_{initial} = \vec{P}_{final}$$

43.1 Case Study: The Immovable Center of Mass

If a system starts at rest ($\vec{P}_{total} = 0$) and has no external forces, its Center of Mass **cannot move**, regardless of what happens inside.

Example 1: The Horse and Carriage
A horse (mass m) walks a distance d along the floor of a carriage (mass M) that is free to roll.

- As the horse moves forward, the carriage must move backward to keep the Center of Mass (X_{CM}) in the exact same spot.
- If the horse moves $+d$ relative to the carriage, the carriage moves a distance ΔX_{car} relative to the ground:

$$\Delta X_{car} = -\frac{md}{M+m}$$

Example 2: The Boat Paradox
A person stands on a boat.
1. **Walking:** As the person walks to the bow, the boat moves backward. The CM of the system stays fixed relative to the water.
2. **Jumping:** The person jumps to shore. They gain forward momentum $+p$, the boat gains backward momentum $-p$. Total is still zero.
3. **Landing:** Friction from the shore stops the person. This is an **external force**.
4. **The Result:** The person stops, but the boat *keeps moving*. The external force acted only on the person, not the boat.

44 Variable Mass Systems: The Rocket

Rockets accelerate by expelling mass backwards. Since the system (Rocket + Exhaust) has no external horizontal forces, momentum is conserved.

44.1 The Rocket Equation

Let M be the rocket mass and v be its velocity. In a time dt, it expels mass dm at exhaust velocity u (relative to the rocket). Using conservation of momentum:

$$Mdv = -udm$$

Integrating this from initial mass M_i to final mass M_f:

$$\Delta v = v_{ex} \ln\left(\frac{M_i}{M_f}\right) \tag{28}$$

This famous equation shows that velocity depends on the exhaust speed and the logarithm of the mass ratio.

45 Collisions

In all collisions (assuming no external forces like friction during the impact), **Total Momentum is Conserved**. However, Kinetic Energy is not always conserved.

45.1 Types of Collisions

1. **Perfectly Inelastic:** The objects stick together after impact.
   - Momentum: Conserved ($p_i = p_f$).
   - Energy: Not Conserved (Significant loss to heat/deformation).
   - Final Velocity: $v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$.
2. **Perfectly Elastic:** The objects bounce off with no loss of energy.
   - Momentum: Conserved.
   - Energy: Conserved ($K_i = K_f$).
   - Requires solving two simultaneous equations to find v_{1f} and v_{2f}.

45.2 Case Study: The Ballistic Pendulum

A bullet (mass m, speed v) hits a wooden block (mass M) hanging from a string. The bullet embeds itself, and the block swings up to height h. This problem requires a two-step analysis.

Step 1: The Collision (Inelastic)
The bullet hits the block. Energy is lost to heat (friction of wood). We must use **Momentum Conservation**:

$$mv = (M + m)V_{\text{start}} \implies V_{\text{start}} = \frac{m}{M + m}v$$

Step 2: The Swing (Conservative)
The block swings up. Only gravity acts (conservative). We use **Energy Conservation**:

$$\frac{1}{2}(M + m)V_{\text{start}}^2 = (M + m)gh$$

Conclusion: By measuring the height h, we can work backward to find the bullet's speed v.

$$v = \left(\frac{M + m}{m}\right)\sqrt{2gh}$$

Warning: You cannot equate initial kinetic energy of the bullet to the final potential energy ($\frac{1}{2}mv^2 \neq (M + m)gh$) because energy was dissipated during the collision step.

46 Dynamics of Rigid Bodies: Translation vs. Rotation

46.1 Rigid Body: Definition

A **rigid body** is an object in which the distance between any two points remains constant during motion. Mathematically, if points A and B are on the body, then

$$|AB| = \text{constant}.$$

Examples: a coin, a eraser, a meter stick. Real objects can bend, but rigid bodies are an idealization.

46.1.1 Point particle vs. extended (rigid) body

A *point particle* is an ideal object with no spatial extent. Its configuration requires only a single vector position $\mathbf{x}(t)$; the notion of orientation or internal deformation is meaningless. A point particle cannot "rotate" by itself because rotation is an operation on spatially extended objects.

An *extended rigid body* is an idealized object whose internal distances are fixed: for any two material points i, j with position vectors $\mathbf{r}_i(t)$ and $\mathbf{r}_j(t)$ measured in an inertial frame,

$$\|\mathbf{r}_i(t) - \mathbf{r}_j(t)\| = \text{constant (independent of } t).$$

This is the mathematical definition of rigidity we adopt.

46.2 Translation

A rigid body undergoes **pure translation** if every point of the body moves the same distance, in the same direction, at the same time. There is no change in orientation.

Formally, if $\vec{r}_i(t)$ is the position of point i in the body, and $\vec{r}_{\text{CM}}(t)$ is the position of the center of mass, then:

$$\vec{r}_i(t) = \vec{r}_{\text{CM}}(t) + \text{constant vector}.$$

Thus, during translation,

$$\frac{d\vec{r}_i}{dt} = \frac{d\vec{r}_{\text{CM}}}{dt}$$

meaning all points have the same velocity.

A point mass can only translate, because it has no internal orientation and therefore no meaningful rotation.

46.3 Rotation

Rotation is a motion in parts of the body move with different velocities around a fixed axis or around the center of mass that arise from a common angular velocity vector $\boldsymbol{\omega}(t)$. For a rigid body rotating about some origin (or about its center of mass), the instantaneous velocity of a material point located at position vector $\mathbf{r}$ *relative to the chosen origin* is

$$\mathbf{v}(\mathbf{r}, t) = \boldsymbol{\omega}(t) \times \mathbf{r}. \tag{29}$$

This formula implies that points at distance $r_\perp$ from the rotation axis have speed $v = \omega r_\perp$ and directions tangent to circles about the axis.

Note: Rotation alone means the center of mass (or chosen reference point) is held fixed ($\mathbf{V}_{\text{cm}} = 0$). A rigid body undergoes **rotation** if different . The orientation of the body changes with time.

The state of the body is described by the angle $\theta(t)$ it makes with a reference axis.

important theorem:

> Any motion of a rigid body from one instant to the next can be built by a translation followed by a rotation.

Translation tells you *where* the body is. Rotation tells you *how* the body is oriented.

46.4 Fixing One Point in the Body

To isolate pure rotation, imagine fixing one point of the body so it cannot translate. Then the only possible motion is rotation around that fixed point.

For a rigid body in three dimensions, the body can rotate in infinitely many ways. To simplify, we restrict motion to the 2D plane.

46.5 Analogy Between Translation and Rotation

Translation	**Rotation**
Position x	Angle θ
Velocity $\dot{x}$	Angular velocity $\dot{\theta} = \omega$
Mass m	Moment of inertia I
Force F	Torque τ
$F = ma$	$\tau = I\alpha$

The entire theory of rotation parallels the theory of translation.

46.6 Why No Rotation in One Dimension?

In 1D there is no place to rotate rotations require at least two dimensions. Thus, rigid body rotation becomes meaningful starting in 2D.

Summary

- A rigid body maintains fixed distances between all points.
- Its motion can be decomposed into translation + rotation.
- For now, we study pure rotation by fixing one point.
- In 2D, orientation is fully described by an angle $\theta(t)$.
- Rotation is mathematically analogous to 1D translation.

47 Measuring Angles: Degrees vs. Radians

In classical mechanics, we measure angles in **Radians** rather than degrees.

- 1 Revolution = $360^\circ = 2\pi$ Radians.

Why do physicists prefer radians? Because in calculus, formulas like $\frac{d}{dx}(\sin x) = \cos x$ are only true if x is in radians. If you use degrees, messy constants $(180/\pi)$ appear everywhere.

47.1 Aside: The Quantum "360 Degree" Paradox

in everyday language, "doing a 360" means returning to where you started. In Quantum Mechanics, this is not always true.

Particles are divided into two categories based on how they behave under rotation:

1. **Bosons (e.g., Photons):** Behave like classical objects.

$$\text{Rotate } 360^\circ \implies \text{State returns to original } (+1).$$

2. **Fermions (e.g., Electrons):** Behave like a "twisted" system (Spin 1/2).

$$\text{Rotate } 360^\circ \implies \text{State becomes negative } (-1).$$

$$\text{Rotate } 720^\circ \implies \text{State returns to original } (+1).$$

Physical Analogy (The Plate Trick):
Imagine holding a plate flat on your palm. If you rotate your hand 360° by passing it under your arm, the plate is flat, but your arm is twisted. You must rotate it *another* 360° to untwist your arm. Fermions behave like this "arm-plate" system; they require two full turns to reset their internal mathematical state.

48 Describing Rotational Motion

When a rigid body rotates, every point on the body moves in a circle. However, different points move at different speeds. To describe the motion of the body as a whole, we need new variables.

48.1 Why Radians?

In elementary geometry, we use degrees (360°). In physics, we use **Radians** because they simplify the relationship between angle and distance.

- **Definition:** An angle θ in radians is defined as the arc length s divided by the radius r.
$$\theta = \frac{s}{r} \implies s = r\theta$$
- A full circle is 2π radians.

If we used degrees, we would carry a factor of $\frac{\pi}{180}$ in every equation. Radians are the "natural" unit of the universe.

48.2 Angular Velocity (ω)

Just as linear velocity v is the rate of change of position (dx/dt), **Angular Velocity** ω is the rate of change of angle:

$$\omega = \frac{d\theta}{dt} \quad \text{(units: rad/s)} \tag{30}$$

48.3 The Critical Distinction: ω vs. v

It is vital to distinguish between the rotation of the body and the speed of a point on the body.

- **Angular Velocity (ω):** This is a property of the **Entire Body**. Every point on a spinning wheel rotates through the same angle in the same time. Therefore, ω is the same for all points.

- **Tangential Velocity (v_T):** This describes how fast a specific point is moving through space (meters per second). It depends on how far the point is from the center (r).

$$v_T = r\omega \tag{31}$$

Physical Implication: Consider a merry-go-round spinning at constant ω.

1. **Near the Center ($r \approx 0$):** You are barely moving ($v \approx 0$).
2. **At the Edge (large r):** You are moving very fast (v is large).

Thus, while the body is "rigid" and moves together angularly, the linear speeds of its parts vary linearly with radius.

49 Rotational Dynamics and Acceleration

Just as linear velocity can change, angular velocity can change. This introduces the concept of angular acceleration.

49.1 Angular Acceleration (α)

The rate of change of angular velocity is defined as angular acceleration:

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} \quad \text{(units: rad/s}^2\text{)} \tag{32}$$

49.2 Tangential vs. Radial Acceleration

A point on a rotating body at radius r experiences two distinct types of acceleration. It is crucial not to confuse them.

1. **Radial (Centripetal) Acceleration (a_R):**
   - Always present if the body is rotating ($\omega \neq 0$).
   - Points toward the center.
   - Magnitude: $a_R = \omega^2 r$.
2. **Tangential Acceleration (a_T):**
   - Present only if the rotation speed is changing ($\alpha \neq 0$).
   - Points tangent to the circle (along the rim).

- Magnitude: $a_T = r\alpha$.

Example (The Bug on a Saw): If a bug clings to the edge of a saw blade that is speeding up, it must hold on against both the centrifugal tendency (radial) and the jerk of speeding up (tangential).

49.3 The Great Analogy: Linear vs. Rotational

Since the mathematical definitions of x, v, a and θ, ω, α are identical, the kinematic equations we learned in Chapter 1 apply directly to rotation (assuming constant acceleration).

Linear Motion (1D)	**Rotational Motion (Fixed Axis)**
Position x Velocity $v = dx/dt$ Acceleration $a = dv/dt$	Angle θ (rad) Angular Velocity $\omega = d\theta/dt$ Angular Acceleration $\alpha = d\omega/dt$
$v = v_0 + at$ $x = x_0 + v_0 t + \frac{1}{2}at^2$ $v^2 = v_0^2 + 2a(x - x_0)$	$\omega = \omega_0 + \alpha t$ $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$ $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$

50 Dynamics of Rigid Bodies: Moment of Inertia

We now translate Newton's laws into the language of rotation.

50.1 Moment of Inertia (I)

Consider a rigid body made of masses $m_1, m_2, \ldots$ at distances $r_1, r_2, \ldots$ from the axis of rotation. The total kinetic energy is the sum of individual kinetic energies:

$$K = \sum \frac{1}{2} m_i v_i^2$$

Since $v_i = r_i \omega$ and ω is the same for all points:

$$K = \sum \frac{1}{2} m_i (r_i \omega)^2 = \frac{1}{2} \left(\sum m_i r_i^2 \right) \omega^2$$

We define the term in parentheses as the **Moment of Inertia**:

$$I = \sum m_i r_i^2 \quad (\text{units: kg} \cdot \text{m}^2) \tag{33}$$

Thus, the rotational kinetic energy is:

$$K_{rot} = \frac{1}{2} I \omega^2 \tag{34}$$

Physical Meaning:
I is the rotational analog of mass. It measures resistance to rotational acceleration. Mass further from the axis contributes much more (r^2) to I than mass near the axis.

51 Dynamics of Rigid Bodies: Torque and Work

Just as Force causes linear acceleration, **Torque** causes angular acceleration. However, unlike force, torque depends on *where* and *how* you push.

51.1 The Definition of Torque (τ)

Consider a door. To rotate it efficiently, you push as far from the hinges as possible and perpendicular to the surface.

- **Radial Force ($F\cos\theta$):** Pulls the object away from the pivot. This does not cause rotation (it just stresses the hinges).
- **Tangential Force ($F\sin\theta$):** Pushes the object along the circle. This causes rotation.

The Torque is defined as:

$$\tau = rF\sin\theta \tag{35}$$

where:

- r is the distance from the pivot to the point where force is applied.
- θ is the angle between the position vector $\vec{r}$ and the force vector $\vec{F}$.

51.2 Work Done by Torque

In linear mechanics, Work is $dW = Fdx$. In rotational mechanics, the linear displacement dx is related to the angular displacement $d\theta$ by the arc length formula $ds = rd\theta$.

Substituting this into the work definition:

$$dW = F_{\text{tangential}} \cdot ds = (F\sin\theta)(rd\theta)$$

Since $\tau = rF\sin\theta$, we get the fundamental relation for rotational work:

$$W = \int_{\theta_1}^{\theta_2} \tau\, d\theta \tag{36}$$

52 The Parallel Axis Theorem

Calculating the Moment of Inertia (I) for an axis passing through the Center of Mass (CM) is usually straightforward due to symmetry. However, calculating it for an arbitrary axis can be difficult mathematically. The Parallel Axis Theorem provides a simple shortcut.

52.1 Statement of the Theorem

If the moment of inertia through the center of mass is I_{CM}, then the moment of inertia I about any other parallel axis is:

$$I = I_{\text{CM}} + Md^2 \tag{37}$$

where:

- M is the total mass of the rigid body.
- d is the perpendicular distance between the two axes.

Physical Interpretation: This theorem separates rotation into two parts:

1. **Spin:** Rotation of the body about its own center (I_{CM}).
2. **Orbital Motion:** The motion of the entire mass M orbiting around the new pivot point at a radius d (Md^2).

Note that I is always minimized when the axis passes through the center of mass ($d = 0$).

52.2 Example: Rotating a Disk from the Edge

Consider a uniform disk of mass M and radius R.

- **Axis through Center:** By integrating concentric rings, we know:

$$I_{\text{CM}} = \frac{1}{2}MR^2$$

- **Axis through the Edge:** Suppose we pin the disk at its rim (distance $d = R$ from the center). Instead of re-integrating, we use the theorem:

$$I_{\text{edge}} = I_{\text{CM}} + M(R)^2$$

$$I_{\text{edge}} = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$$

This result ($1.5MR^2$) tells us it is three times harder to spin a coin around its edge than to spin it around its center.

52.3 Proof using Vectors

Let the position of a mass element relative to the CM be $\vec{r}_i$ and the position of the new axis relative to the CM be $\vec{d}$. The position relative to the new axis is $\vec{r}_i' = \vec{r}_i + \vec{d}$.

$$I = \sum m_i (\vec{r}_i')^2 = \sum m_i (\vec{r}_i + \vec{d}) \cdot (\vec{r}_i + \vec{d})$$

Expanding the dot product:

$$I = \underbrace{\sum m_i r_i^2}_{I_{\text{CM}}} + \underbrace{\sum m_i d^2}_{Md^2} + \underbrace{2\vec{d} \cdot \sum m_i \vec{r}_i}_{0}$$

The last term vanishes because $\sum m_i \vec{r}_i$ defines the center of mass position. Since our origin is at the CM, this sum is exactly zero.

53 Kinetic Energy of a System of Particles

Just as we decomposed the moment of inertia using the Parallel Axis Theorem, we can decompose the kinetic energy of a moving, rotating system into two independent parts.

53.1 Decomposition of Velocity

Consider a system of particles with masses m_i and velocities $\vec{v}_i$. Let $\vec{V}_{\text{CM}}$ be the velocity of the Center of Mass. The velocity of any particle can be written as:

$$\vec{v}_i = \vec{V}_{\text{CM}} + \vec{v}_{i,\text{rel}}$$

where $\vec{v}_{i,\text{rel}}$ is the velocity of the particle relative to the Center of Mass.

53.2 Calculating Total Kinetic Energy

The total kinetic energy is:

$$K = \sum \frac{1}{2} m_i v_i^2 = \sum \frac{1}{2} m_i (\vec{V}_{\text{CM}} + \vec{v}_{i,\text{rel}}) \cdot (\vec{V}_{\text{CM}} + \vec{v}_{i,\text{rel}})$$

Expanding the dot product gives three terms:

1. **Translational Term:** $\frac{1}{2}(\sum m_i) V_{\text{CM}}^2 = \frac{1}{2} M V_{\text{CM}}^2$.
2. **Relative Motion Term:** $\sum \frac{1}{2} m_i v_{i,\text{rel}}^2$. For a rigid body, the only relative motion allowed is rotation, so this becomes $\frac{1}{2} I_{\text{CM}} \omega^2$.
3. **Cross Term:** $\vec{V}_{\text{CM}} \cdot (\sum m_i \vec{v}_{i,\text{rel}})$.

Why the Cross Term Vanishes:
The term $\sum m_i \vec{v}_{i,\text{rel}}$ is the total momentum of the system measured from the Center of Mass frame. By definition of the Center of Mass, the net momentum in its own frame is zero. Thus, the cross term disappears.

53.3 The Final Result: Translation + Rotation

For a rigid body undergoing both translation and rotation, the total kinetic energy is simply the sum of the translational energy of the center of mass and the rotational energy about the center of mass:

$$K_{\text{total}} = \frac{1}{2}MV_{\text{CM}}^2 + \frac{1}{2}I_{\text{CM}}\omega^2 \tag{38}$$

Physical Significance: This theorem allows us to analyze complex motion (like a thrown baton or a rolling wheel) by treating it as two simple motions added together:

- The motion of a point mass M following a trajectory.
- The spinning of the object around that point.

54 Rolling Motion

Rolling is a combination of translation and rotation. We can analyze it by decomposing the motion into the velocity of the center of mass (V) and the angular velocity (ω) around the center.

54.1 Rolling Without Slipping

Generally, V and ω are independent (e.g., a skidding car has $V \neq 0, \omega = 0$). However, ideal rolling occurs when the point of contact does not slip.

- In one revolution (time T), the wheel rotates 2π radians and travels a distance $2\pi R$.
- Velocity $V = \frac{\text{Distance}}{T} = \frac{2\pi R}{T} = \left(\frac{2\pi}{T}\right) R = \omega R$.

The condition for rolling without slipping is:

$$V = \omega R \tag{39}$$

54.2 Kinetic Energy of a Rolling Wheel

Using the decomposition theorem:

$$K = K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}MV^2 + \frac{1}{2}I_{\text{CM}}\omega^2$$

For a solid disk ($I_{\text{CM}} = \frac{1}{2}MR^2$) rolling without slipping ($\omega = V/R$):

$$K = \frac{1}{2}MV^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\left(\frac{V}{R}\right)^2$$

$$K = \frac{1}{2}MV^2 + \frac{1}{4}MV^2 = \frac{3}{4}MV^2$$

This shows that a rolling object has 50% more energy than a sliding object at the same speed (due to the rotational term).

54.3 The Instantaneous Point of Rotation

We can alternatively view the motion as pure rotation about the **point of contact** with the ground (P). Using the Parallel Axis Theorem, the moment of inertia about the contact point is:

$$I_P = I_{\text{CM}} + MR^2 = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$$

The rotational energy about this point is:

$$K = \frac{1}{2}I_P\omega^2 = \frac{1}{2}\left(\frac{3}{2}MR^2\right)\left(\frac{V}{R}\right)^2 = \frac{3}{4}MV^2$$

This matches the previous result perfectly.

54.4 Velocities of Points on the Tire

Why can we treat the contact point P as a fixed pivot? Because its instantaneous velocity is zero! The velocity of any point is the vector sum of translation (V) and rotation ($\vec{v}_{\text{tan}}$):

- **Top of Tire:** Translation V (Right) + Rotation ωR (Right) $\Rightarrow v_{\text{top}} = V + V = 2V$.
- **Center:** Translation V (Right) + Rotation 0 $\Rightarrow v_{\text{center}} = V$.
- **Bottom (Contact Point):** Translation V (Right) + Rotation ωR (Left) $\Rightarrow v_{\text{bottom}} = V - V = 0$.

Since the bottom point is momentarily at rest relative to the ground, static friction applies, allowing the car to accelerate without burning rubber.

55 Applications of Rotational Dynamics

We now combine the concepts of Moment of Inertia, Rolling Motion, and Torque to solve complex problems.

55.1 Rolling Down an Incline

Consider an object (mass M, radius R, moment of inertia $I = cMR^2$) rolling down a ramp of height h.

- **Energy Conservation:** Potential energy converts to both translational and rotational kinetic energy.

$$mgh = K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

- **Rolling Condition:** $\omega = v/R$.

$$mgh = \frac{1}{2}Mv^2 + \frac{1}{2}(cMR^2)\left(\frac{v}{R}\right)^2$$

$$mgh = \frac{1}{2}Mv^2(1+c)$$

- **Final Velocity:**

$$v = \sqrt{\frac{2gh}{1+c}} \tag{40}$$

Conclusion: The velocity is lower than that of a sliding block ($v = \sqrt{2gh}$) by a factor depending on shape (c).

- Solid Sphere ($c = 2/5$): $v = \sqrt{10/7\, gh} \approx 1.2\sqrt{gh}$
- Solid Disk ($c = 1/2$): $v = \sqrt{4/3\, gh} \approx 1.15\sqrt{gh}$
- Hollow Ring ($c = 1$): $v = \sqrt{gh} \approx 1.0\sqrt{gh}$ (Slowest)

55.2 Torque Dynamics: The Merry-Go-Round

Consider a disk (mass M, radius R) mounted on a pivot. We attach two rockets on opposite sides, each providing thrust F.

- **Forces:** The two forces point in opposite directions, so $F_{\text{net}} = 0$. The center of mass does not move.

- **Torques:** Both forces create torque in the same direction (e.g., counter-clockwise).

$$\tau_{\text{net}} = (F \cdot R) + (F \cdot R) = 2FR$$

- **Angular Acceleration:**

$$\alpha = \frac{\tau_{\text{net}}}{I} = \frac{2FR}{\frac{1}{2}MR^2} = \frac{4F}{MR}$$

If we add a passenger (mass m) at radius r, the moment of inertia increases to $I_{\text{new}} = \frac{1}{2}MR^2 + mr^2$, reducing the acceleration.

55.3 The Massive Pulley

Consider a mass m hanging from a string wrapped around a massive pulley (disk of mass M, radius R). Unlike previous problems, the pulley has inertia.

1. Equation for the Hanging Mass (Linear): Gravity pulls down, Tension T pulls up.

$$mg - T = ma$$

2. Equation for the Pulley (Rotational): The tension T creates a torque on the pulley.

$$\tau = T \cdot R = I\alpha$$

Since the string does not slip, $\alpha = a/R$.

$$T \cdot R = \left(\frac{1}{2}MR^2\right)\left(\frac{a}{R}\right) \implies T = \frac{1}{2}Ma$$

3. Solving for Acceleration: Substitute T into the first equation:

$$mg - \frac{1}{2}Ma = ma \implies mg = a(m + M/2)$$

$$a = \frac{mg}{m + M/2} = \frac{g}{1 + M/2m} \tag{41}$$

The acceleration is less than g. The inertia of the heavy pulley slows down the fall.

56 Physical Meaning

explains the ideas behind the equations in plain physical language, without adding new mathematics.

56.1 Why rolling objects move slower than sliding ones

When an object rolls down an incline, gravity does *one job* but the energy is *split into two parts*:

- Translational motion of the center of mass (moving forward).
- Rotational motion around the center of mass (spinning).

A sliding block uses *all* of its gravitational potential energy to increase its forward speed. A rolling object must also spin, so part of the energy is "spent" on rotation.

This is why rolling objects always reach the bottom with a *smaller speed* than a sliding block dropped from the same height.

56.2 Meaning of the shape factor c

The constant c in $I = cMR^2$ tells us how the mass is distributed:

- Small c: mass is closer to the center $\rightarrow$ easier to spin.
- Large c: mass is farther from the center $\rightarrow$ harder to spin.

Objects that are harder to spin steal more energy for rotation, leaving less energy for forward motion. That is why:

- A solid sphere (small c) rolls fastest.
- A hollow ring (large c) rolls slowest.

56.3 Why zero net force can still cause rotation

In the merry-go-round example, the two forces cancel as *forces*, so the center of mass does not move.

However, each force is applied at a distance from the center, so each one produces a *turning effect* (torque).

Torque depends on:

- The force magnitude.
- How far from the center the force is applied.

Even when the total force is zero, the torques can add up and cause angular acceleration. This shows that:

Force controls translation, torque controls rotation.

56.4 Why adding a passenger slows rotation

Adding a passenger increases the moment of inertia because mass is added at some distance from the axis.

Physically:

- Larger moment of inertia means more resistance to changes in rotation.
- The same torque now has to "spin more mass".

This is why the angular acceleration decreases when extra mass is added far from the center.

56.5 Physical meaning of the massive pulley

In the pulley system, gravity does not only accelerate the falling mass.

Some of the gravitational energy is used to:

- Move the mass downward.
- Spin the pulley.

Because the pulley has inertia, it resists rotation. The tension in the string must both:

- Pull the mass.
- Produce torque to rotate the pulley.

As a result, the mass falls slower than it would if the pulley were massless.

56.6 Key takeaway

Whenever rotation is involved:

- Energy is shared between translation and rotation.
- Torque, not force, controls rotational motion.
- Moment of inertia plays the same role in rotation as mass does in translation.

57 Conservation of Angular Momentum

Just as linear momentum is conserved in the absence of external forces, angular momentum is conserved in the absence of external torques.

$$\text{If } \tau_{\text{ext}} = 0, \quad \text{then } L_{\text{total}} = \text{Constant} \tag{42}$$

57.1 Example 1: The Rotating Disks

Consider a disk with moment of inertia I_1 spinning at ω_1. A second non-rotating disk I_2 is dropped onto it. Friction acts between them until they spin together.

- **Before:** $L_i = I_1\omega_1 + I_2(0) = I_1\omega_1$.
- **After:** They rotate as one unit with inertia $(I_1 + I_2)$ at speed ω_f.

$$L_f = (I_1 + I_2)\omega_f$$

- **Conservation:** $I_1\omega_1 = (I_1 + I_2)\omega_f \implies \omega_f = \omega_1 \frac{I_1}{I_1+I_2}$.

The final speed is lower because the rotational mass increased.

57.2 Example 2: Angular Momentum of a Linear Particle

A particle of mass m moving in a straight line with velocity v at a distance R (impact parameter) from an axis has angular momentum relative to that axis.

$$L = mvR \tag{43}$$

Application: If a piece of putty is shot at the rim of a stationary merry-go-round, we treat the putty as having initial angular momentum $L = mvR$. After impact, the system (Disk + Putty) spins with that total L.

57.3 The Ice Skater Effect

Consider a skater spinning with arms outstretched (I_{out}) at speed ω_1. She pulls her arms in (I_{in}). Since $I_{\text{in}} < I_{\text{out}}$ (mass is closer to the axis), and $L = I\omega$ is conserved:

$$I_{\text{out}}\omega_1 = I_{\text{in}}\omega_2 \implies \omega_2 = \omega_1 \left(\frac{I_{\text{out}}}{I_{\text{in}}}\right)$$

Since the ratio is greater than 1, the skater spins faster.

Energy Analysis: Does energy stay the same?

$$K_1 = \frac{1}{2} I_{\text{out}} \omega_1^2 \quad \text{vs} \quad K_2 = \frac{1}{2} I_{\text{in}} \omega_2^2$$

Substituting ω_2:

$$K_2 = \frac{1}{2} I_{\text{in}} \left(\omega_1 \frac{I_{\text{out}}}{I_{\text{in}}} \right)^2 = \left(\frac{1}{2} I_{\text{out}} \omega_1^2 \right) \frac{I_{\text{out}}}{I_{\text{in}}} = K_1 \left(\frac{I_{\text{out}}}{I_{\text{in}}} \right)$$

Since $I_{\text{out}} > I_{\text{in}}$, $^{**}K_2 > K_1{}^{**}$. The kinetic energy **increases**. This extra energy comes from the **Work** done by the skater's muscles to pull her arms inward against the centrifugal tendency.

58 Static Equilibrium

previously studied dynamics, where forces cause linear acceleration ($\vec{F} = m\vec{a}$) and torques cause angular acceleration ($\tau = I\alpha$). We now consider the special case where the object remains completely at rest.

58.1 Definition

An object is in **Static Equilibrium** if it has no linear velocity and no angular velocity. For a rigid body in two dimensions (the xy-plane), this requires three conditions to be met simultaneously:

1. **No Horizontal Acceleration:** The sum of all forces in the x-direction must be zero.
$$\sum_i F_{i,x} = 0$$
2. **No Vertical Acceleration:** The sum of all forces in the y-direction must be zero.
$$\sum_i F_{i,y} = 0$$
3. **No Rotational Acceleration:** The sum of all torques about *any* pivot point must be zero.
$$\sum_i \tau_i = 0$$

58.2 Why Force Balance is Not Enough

Consider a rod where we apply an upward force F on the left end and an equal downward force F on the right end.

- The net force is zero: $\sum F = F - F = 0$.
- The Center of Mass does not move.
- However, the rod will **rotate**.

Thus, $\sum \vec{F} = 0$ ensures translational equilibrium, but $\sum \tau = 0$ is required for rotational equilibrium.

58.3 Example: The Seesaw

Consider a plank supported by a fulcrum (pivot). Two forces F_1 and F_2 (weights) act downwards at distances x_1 and x_2 from the pivot. The pivot exerts an upward normal force N.

1. Force Equation (y-direction):

$$\sum F_y = N - F_1 - F_2 = 0 \implies N = F_1 + F_2$$

This tells us the pivot supports the total weight.

2. Torque Equation: We calculate torque relative to the pivot point.

- F_1 creates a counter-clockwise (positive) torque: $\tau_1 = +F_1 x_1$.
- F_2 creates a clockwise (negative) torque: $\tau_2 = -F_2 x_2$.
- N creates zero torque because it acts *at* the pivot (distance = 0).

For equilibrium:

$$\sum \tau = F_1 x_1 - F_2 x_2 = 0 \implies F_1 x_1 = F_2 x_2$$

This allows us to solve for an unknown force (e.g., $F_2 = F_1 \frac{x_1}{x_2}$).

59 Torque Due to Gravity

Gravity acts on every particle of an extended object. To calculate the total torque about a pivot, we sum the torques on individual mass elements dm located at position x.

$$\tau_{\text{gravity}} = \int x(g\,dm) = g \int x\,dm$$

Recall the definition of the Center of Mass (x_{CM}):

$$x_{\text{CM}} = \frac{1}{M} \int x\,dm \implies \int x\,dm = M x_{\text{CM}}$$

Substituting this back:

$$\tau_{\text{gravity}} = Mgx_{\text{CM}} \tag{44}$$

Conclusion: For the purpose of calculating torque, we can treat the entire weight of the object (Mg) as acting at a single point: the Center of Mass. This assumes g is constant across the object.

60 Strategy for Solving Static Problems

A rigid body in equilibrium has two unknown reaction forces at the pivot (horizontal H and vertical V). To solve for other forces without getting bogged down in algebra:

- **Pick the Pivot Point Strategically:** Calculate torques about the point where the unknown pivot forces act.
- Since the lever arm $r = 0$ for these forces, their torque is zero ($\tau = 0$).
- This eliminates H and V from the equation, allowing you to solve for the remaining unknowns directly.

60.1 Example: Rod Supported by a Cable

Consider a uniform rod of length L and mass M, hinged at a wall and supported by a cable at the far end making an angle θ with the rod.

1. Torque Equation (about the hinge): The hinge forces (H, V) create zero torque.

- Torque from Gravity (Clockwise): acts at $L/2$.

$$\tau_g = -Mg\left(\frac{L}{2}\right)$$

- Torque from Tension (Counter-Clockwise): acts at L. Only the perpendicular component $T\sin\theta$ causes rotation.

$$\tau_T = +(T\sin\theta)L$$

Equilibrium ($\sum \tau = 0$):

$$Mg\frac{L}{2} = TL\sin\theta \implies T = \frac{Mg}{2\sin\theta}$$

2. Force Equations (to find hinge forces): Now that T is known, we use $\sum F_x = 0$ and $\sum F_y = 0$.

- Horizontal: The cable pulls left ($T\cos\theta$). The hinge must push right (H).

$$H - T\cos\theta = 0 \implies H = T\cos\theta = \frac{Mg}{2}\cot\theta$$

- Vertical: The cable pulls up ($T\sin\theta$) and hinge pushes up (V). Gravity pulls down.

$$V + T\sin\theta - Mg = 0$$

Substituting $T\sin\theta = Mg/2$:

$$V + \frac{Mg}{2} - Mg = 0 \implies V = \frac{Mg}{2}$$

Result: The hinge supports exactly half the weight of the rod, and the vertical component of the tension supports the other half.

61 The Ladder Problem

This is the classic application of static equilibrium. We calculate the minimum angle θ required for a ladder to lean against a wall without slipping.

61.1 Problem Setup

A uniform ladder of mass M and length L leans against a frictionless wall at angle θ. The floor has a coefficient of static friction μ_s.

Forces Analysis:

- **Gravity (Mg):** Acts down at the center of mass ($L/2$).
- **Wall Force (W):** Normal force from the wall, acts horizontally to the left.
- **Floor Normal Force (N):** Acts vertically up.
- **Friction (f):** Acts horizontally to the right (opposing the tendency to slip).

61.2 Step 1: Force Equilibrium

The sum of forces must be zero.

$$\sum F_y = 0 \implies N - Mg = 0 \implies \boxed{N = Mg}$$
$$\sum F_x = 0 \implies f - W = 0 \implies \boxed{f = W}$$

61.3 Step 2: Torque Equilibrium

We choose the **bottom of the ladder** as the pivot point. This eliminates N and f from the torque equation because their lever arm is zero.

- **Torque from Wall (W):** Counter-clockwise.

$$\tau_W = W \cdot L \cdot \sin(\theta)$$

- **Torque from Gravity (Mg):** Clockwise. The angle between the vertical force and the ladder is $90° - \theta$.

$$\tau_g = Mg \cdot \frac{L}{2} \cdot \sin(90° - \theta) = Mg\frac{L}{2}\cos(\theta)$$

Equating magnitudes ($\sum \tau = 0$):

$$WL\sin\theta = \frac{MgL}{2}\cos\theta$$

Solving for W:

$$W = \frac{Mg\cos\theta}{2\sin\theta} = \frac{Mg}{2\tan\theta}$$

61.4 Step 3: The Slip Condition

For the ladder to stay static, the required friction ($f = W$) must not exceed the maximum possible static friction ($f_{max} = \mu_s N$).

$$f \leq \mu_s N$$

Substitute the derived values:

$$\frac{Mg}{2\tan\theta} \leq \mu_s(Mg)$$

Canceling Mg:

$$\frac{1}{2\tan\theta} \leq \mu_s \implies \tan\theta \geq \frac{1}{2\mu_s}$$

Physical Meaning: To prevent slipping, the angle θ must be steep enough. If the floor is slippery (μ_s is low), $\tan\theta$ must be very high (the ladder must be almost vertical). If the floor is rough (μ_s is high), the ladder can be placed at a shallower angle.

62 Rigid Body Dynamics in 3D

When rotations occur in 3D space, we must treat Torque and Angular Momentum as full vectors, not just scalars with a sign. This requires the use of the Vector Cross Product.

62.1 The Cross Product

The cross product of two vectors $\vec{A}$ and $\vec{B}$ is a third vector $\vec{C} = \vec{A} \times \vec{B}$ with the following properties:

- **Magnitude:** $|\vec{C}| = |A||B| \sin\theta$, where θ is the angle between them.
- **Direction:** Perpendicular to the plane formed by $\vec{A}$ and $\vec{B}$, determined by the **Right-Hand Rule**.
- **Order Matters:** $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$.

62.2 Torque and Angular Momentum Vectors

The vector definitions for rotational quantities are:

$$\vec{\tau} = \vec{r} \times \vec{F} \tag{45}$$

$$\vec{L} = \vec{r} \times \vec{p} \tag{46}$$

The fundamental law of dynamics becomes:

$$\vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt} \tag{47}$$

This means the external torque dictates the *velocity of the tip* of the angular momentum vector.

62.3 The Gyroscope

Consider a spinning wheel of mass M supported at a pivot a distance d from its center of mass.

- **Angular Momentum:** The wheel spins rapidly with angular velocity ω_s. It has a large angular momentum $\vec{L}$ pointing along the axle.
- **Torque:** Gravity (Mg) acts downwards. The torque about the pivot is:

  $$\vec{\tau} = \vec{r} \times \vec{F}$$

  Since $\vec{r}$ is horizontal and $\vec{F}$ is vertical, the torque vector is **horizontal** and perpendicular to the axle.

Why doesn't it fall?
The torque $\vec{\tau}$ causes a change in angular momentum $d\vec{L}$. Since $\vec{\tau}$ is perpendicular to $\vec{L}$,

it does not change the magnitude of $\vec{L}$ (the spin speed), but only its **direction**. The tip of the vector $\vec{L}$ moves in a circle. This horizontal rotation is called **Precession**.

Calculating Precession Frequency (Ω_p): In a small time dt, the change in angular momentum is $dL = \tau dt$. The angle of rotation is $d\phi \approx \frac{dL}{L} = \frac{\tau dt}{L}$. The rate of precession is:

$$\Omega_p = \frac{d\phi}{dt} = \frac{\tau}{L} = \frac{Mgd}{I\omega_s} \tag{48}$$

This explains why a faster spin (ω_s) leads to slower precession.

63 Introduction to Relativity

63.1 The Principle of Relativity (Before Einstein)

Relativity is not a new idea introduced by Einstein. Long before him, Galileo and Newton understood a fundamental fact about motion:

> *Uniform motion cannot be detected by experiments performed entirely within the system.*

This means that if an observer is moving with a constant velocity (no acceleration), all mechanical experiments behave exactly as they would if the observer were at rest.

63.2 Inertial Frames

An **inertial frame of reference** is defined as a frame in which Newton's laws hold. In particular:

- A body at rest remains at rest.
- A body in motion continues in uniform straight-line motion unless acted upon by a force.

Observers in inertial frames cannot detect their own uniform velocity without referencing something outside their frame.

63.3 Train Thought Experiment

Consider an observer inside a closed train moving along the x-axis with constant velocity u.

- All blinds are closed.
- No external reference is visible.
- Mechanical experiments (throwing balls, collisions, falling objects) behave normally.

If the train moves with constant velocity, the observer cannot determine whether the train is at rest or moving uniformly. However, if the train accelerates or decelerates, the observer immediately detects it through inertial forces.

63.4 Uniform Motion vs. Acceleration

- **Uniform velocity** is *relative*: different observers may disagree on who is moving.
- **Acceleration** is *absolute*: it produces observable physical effects (forces, discomfort, deformation).

Thus, only inertial frames (non-accelerating frames) are equivalent in Newtonian relativity.

63.5 Events and Space-Time

An **event** is something that occurs at a specific place and time. In one spatial dimension, an event is labeled by:

$$(x, t)$$

This concept of space-time predates Einstein. The novelty of special relativity lies not in defining events, but in redefining how different observers relate their coordinates.

63.6 Two Reference Frames

Let:

- S be a reference frame at rest.
- S' be a frame moving with constant velocity u along the x-axis.

When the origins of S and S' coincide, both observers set their clocks to zero:

$$x = x' = 0, \quad t = 0$$

This coincidence is a single event, agreed upon by both observers.

63.7 Galilean Transformation

For an event observed at position x and time t in frame S, the position in frame S' is:

$$x' = x - ut \tag{49}$$

Time is the same for both observers:

$$t' = t$$

This transformation preserves Newton's laws and expresses the classical principle of relativity.

63.8 Physical Meaning

The Galilean transformation shows that:

- Observers in uniform relative motion disagree on position.
- They agree on time.
- The laws of mechanics remain unchanged.

This is the Newtonian foundation upon which Einstein later builds special relativity.

64 The Crisis of Classical Physics

Newtonian mechanics worked perfectly for 200 years. It obeyed the **Principle of Relativity**: laws of physics are the same for all observers in uniform motion. However, the discovery of electromagnetism and the speed of light shattered this harmony.

64.1 Galilean Transformation

Consider two observers: S (stationary) and S' (moving to the right with velocity u). If an event occurs at (x, t), the coordinates in the moving frame are:

$$x' = x - ut \tag{50}$$

$$t' = t \quad \text{(Universal Time)} \tag{51}$$

Taking derivatives gives the velocity addition rule:

$$v' = v - u$$

Taking derivatives again gives acceleration:

$$a' = a$$

Since $F = ma$ depends on acceleration, Newton's laws are invariant. Both observers see the same physics.

64.2 The Problem with Light

Maxwell's equations showed that light is an electromagnetic wave traveling at speed c. According to Galilean relativity, if you move at speed u towards a light beam, you should measure its speed as:

$$c' = c + u$$

The Experiment: Michelson and Morley tried to measure this change in speed as the Earth moved through the hypothetical "Ether." **The Result:** They found **zero change**. Light always travels at c, regardless of the observer's speed.

$$c' = c$$

64.3 Einstein's Postulate

We faced a contradiction:

- **Newton:** Velocities add ($v' = v - u$).
- **Maxwell:** Light speed is constant ($c' = c$).

Einstein resolved this by discarding the Galilean transformation (specifically, the idea of universal time). He postulated:

1. **Principle of Relativity:** The laws of physics are the same in all inertial frames (no "absolute rest").
2. **Constancy of Light:** The speed of light c is the same for all observers, independent of the motion of the source.

This requires a new transformation rule (Lorentz Transformation) where time itself (t) changes depending on your speed.

65 Derivation of the Lorentz Transformation

Einstein resolved the contradiction between Newton and Maxwell by proposing that space and time are not absolute. We seek a transformation relating coordinates (x, t) in frame S to (x', t') in frame S' (moving at velocity u) that preserves the speed of light.

65.1 The Assumptions

1. **Linearity:** The transformation must be linear to preserve the law of inertia (straight lines remain straight).
2. **Symmetry:** If observer S sees S' moving at $+u$, then S' sees S moving at $-u$. The scaling factor γ must be the same for both.

We propose a modification to the Galilean transformation:

$$x' = \gamma(x - ut) \tag{52}$$
$$x = \gamma(x' + ut') \tag{53}$$

Here, γ is an unknown function of velocity u.

65.2 Enforcing the Light Postulate

Consider a light pulse emitted at the origin when $t = t' = 0$.

- In frame S, the pulse position is $x = ct$.
- In frame S', the pulse position is $x' = ct'$.

Substitute these into our equations:

$$ct' = \gamma(ct - ut) = \gamma t(c - u)$$
$$ct = \gamma(ct' + ut') = \gamma t'(c + u)$$

Multiply the two equations:

$$c^2 tt' = \gamma^2 tt'(c - u)(c + u)$$

Cancel tt' and use $(c - u)(c + u) = c^2 - u^2$:

$$c^2 = \gamma^2(c^2 - u^2)$$

Solving for γ:

$$\gamma^2 = \frac{c^2}{c^2 - u^2} = \frac{1}{1 - u^2/c^2}$$
$$\gamma = \frac{1}{\sqrt{1 - u^2/c^2}} \tag{54}$$

This is the famous **Lorentz Factor**. Note that $\gamma \geq 1$.

65.3 The Transformation Equations

Having found γ, we can solve the system for t'. The resulting set of equations is the **Lorentz Transformation**:

$$x' = \gamma(x - ut) \tag{55}$$

$$t' = \gamma\left(t - \frac{u}{c^2}x\right) \tag{56}$$

Physical Implications:

- **Time is Relative:** The time t' depends on your position x. Two events simultaneous in S ($\Delta t = 0$) are not necessarily simultaneous in S'.
- **The Cosmic Speed Limit:** If $u \geq c$, γ becomes undefined or imaginary. Massive objects cannot reach the speed of light.
- **Low Speed Limit:** If $u \ll c$, then $\gamma \approx 1$ and $u/c^2 \approx 0$. The equations reduce to the Galilean transformation $x' = x - ut$ and $t' = t$. Einstein reduces to Newton at low speeds.

66 Relativistic Velocity Addition

The Lorentz transformations imply that velocities do not simply add up ($v \neq u + w$). We derive the correct relativistic formula to ensure that nothing exceeds the speed of light.

66.1 Derivation

Let u be the speed of a train relative to the ground (S). Let w be the speed of a bullet relative to the train (S'), so $w = dx'/dt'$. We want to find v, the speed of the bullet relative to the ground ($v = dx/dt$).

From the Lorentz equations:

$$dx = \gamma(dx' + udt')$$

$$dt = \gamma(dt' + \frac{u}{c^2}dx')$$

Dividing dx by dt:

$$v = \frac{dx}{dt} = \frac{\gamma(dx' + udt')}{\gamma(dt' + \frac{u}{c^2}dx')}$$

Cancel γ and divide the numerator and denominator by dt':

$$v = \frac{\frac{dx'}{dt'} + u}{1 + \frac{u}{c^2}\frac{dx'}{dt'}}$$

Substituting $w = dx'/dt'$ gives the **Velocity Addition Formula**:

$$v = \frac{w + u}{1 + \frac{wu}{c^2}} \tag{57}$$

66.2 Why Nothing Exceeds c

Let's try to break the light barrier. Suppose a rocket moves at $u = 0.75c$ and fires a missile forward at $w = 0.75c$.

- **Galilean Prediction:** $v = 0.75c + 0.75c = 1.5c$ (Faster than light).
- **Relativistic Prediction:**

$$v = \frac{0.75c + 0.75c}{1 + \frac{(0.75c)(0.75c)}{c^2}} = \frac{1.5c}{1 + 0.5625} = \frac{1.5c}{1.5625} \approx 0.96c$$

The result is still less than c. The denominator "penalizes" high speeds, ensuring the total never reaches the speed of light.

66.3 Consistency with Light

If the "bullet" is actually a beam of light, then $w = c$. The ground observer should also see it move at c.

$$v = \frac{c + u}{1 + \frac{cu}{c^2}} = \frac{c + u}{1 + \frac{u}{c}} = \frac{c(1 + \frac{u}{c})}{1 + \frac{u}{c}} = c$$

The formula works perfectly: the speed of light is absolute.

67 Consequences of Special Relativity

The Lorentz transformations imply that measurements of time and space depend on the observer's motion. We define two key effects: Time Dilation and Length Contraction.

67.1 Relativity of Simultaneity

Events that are simultaneous in one frame are not necessarily simultaneous in another. From the time transformation equation:

$$\Delta t' = \gamma \left(\Delta t - \frac{u}{c^2} \Delta x \right)$$

If two events happen at the same time in frame S ($\Delta t = 0$) but at different locations ($\Delta x \neq 0$), then in frame S':

$$\Delta t' = -\gamma \frac{u}{c^2} \Delta x \neq 0$$

Conclusion: "Now" is a relative concept. There is no universal present moment.

67.2 Time Dilation

Consider a clock at rest in frame S' (the "moving" clock). It measures a time interval Δt_0 (Proper Time) between two ticks at the same location ($\Delta x' = 0$). To an observer in frame S, the time interval Δt is:

$$\Delta t = \gamma \left(\Delta t' + \frac{u}{c^2} \Delta x' \right) = \gamma(\Delta t_0 + 0)$$

$$\Delta t = \gamma \Delta t_0 \tag{58}$$

Since $\gamma \geq 1$, the moving clock runs slower ($\Delta t > \Delta t_0$). This effect applies to all biological and mechanical processes.

67.3 Length Contraction

Consider a rod at rest in frame S' with length $L_0 = \Delta x'$ (Proper Length). An observer in frame S measures the length L by locating both ends *simultaneously* ($\Delta t = 0$). From the Lorentz equation $\Delta x' = \gamma(\Delta x - u\Delta t)$:

$$L_0 = \gamma(L - 0) \implies L = \frac{L_0}{\gamma}$$

$$L = \frac{L_0}{\gamma} \tag{59}$$

Since $\gamma \geq 1$, moving objects appear shorter along the direction of motion.

67.4 Resolving Paradoxes

1. The Twin Paradox: Why isn't the situation symmetric? The traveling twin must accelerate to turn around and return. Acceleration is not relative; the traveler feels forces that the stay-at-home twin does not. The traveler ages less.

2. The Garage Paradox: Does the car fit? A long car moves into a short garage.

- **Garage Frame:** The car shrinks and fits inside for a brief instant.
- **Car Frame:** The garage shrinks. The car does not fit. The front hits the back wall before the rear enters the front door.

Both observers agree on physical outcomes (e.g., if the car crashes), but they disagree on the *timing* of events due to the relativity of simultaneity.

68 Physical Meaning: Time Order, Causality, and Light

The Lorentz transformation shows that space and time mix between observers:

$$\Delta t' = \frac{\Delta t - \frac{u\,\Delta x}{c^2}}{\sqrt{1 - u^2/c^2}}.$$

This means that the time order of two events is not absolute.

Key Idea (Intuition). If two events are separated by a distance Δx and a time Δt, different observers may disagree on which event happened first. The crucial quantity is $c\Delta t$, the distance light can travel in the time between the events.

Causal vs Non-Causal Events.

- If $c\Delta t > \Delta x$, a light signal could travel from the first event to the second. The events can be causally connected, and **all observers agree on their time order**.
- If $c\Delta t < \Delta x$, even light cannot travel between the events in time. The events are not causally connected, and **their time order can be reversed** for some observers moving at speed $u < c$.

Physical Principle. To avoid logical contradictions (cause occurring after effect), relativity demands that no signal, energy, or information travels faster than light. This guarantees that causal events never appear reversed in time for any observer.

Philosophical View. Space and time are not separate absolutes; they form a unified spacetime. Just as rotated observers disagree on spatial coordinates, moving observers disagree on time ordering except where causality, protected by the speed of light, forbids disagreement.

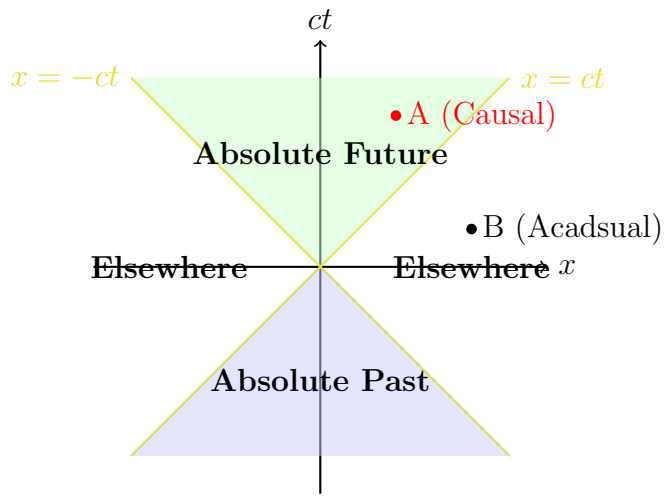
69 The Structure of Spacetime

Special Relativity unifies space and time into a single 4-dimensional continuum. To understand causal relationships, we use **Space-Time Diagrams**.

69.1 The Space-Time Diagram

We plot time on the vertical axis (scaled by c to give it units of length, ct) and space (x) on the horizontal axis.

- A light pulse emitted at the origin travels at $x = ct$. On this graph, light paths are lines at 45°.
- These lines form the **Light Cone**.



69.2 Causality and the Light Cone

The speed of light c is the maximum speed of information. This divides spacetime into three regions relative to an observer at the origin $(0, 0)$:

1. **Absolute Future ($ct > |x|$):** Events inside the upper cone. A signal sent from the origin moving slower than light can reach these points. The observer can influence these events. All observers agree these events happen *after* $t = 0$.
2. **Absolute Past ($ct < -|x|$):** Events inside the lower cone. A signal from these points could have reached the origin. These events could have caused the present moment. All observers agree these happen *before* $t = 0$.
3. **Elsewhere / Spacelike Separated ($|x| > |ct|$):** Events outside the cone. Even light is not fast enough to connect the origin to these points.
   - **No Causality:** Nothing at the origin can affect these events, and vice versa.
   - **Relative Time:** Different observers can disagree on the time order. One might see Event A happen before B, while another sees B before A. This causes no paradoxes because the events are not causally connected.

69.3 The Spacetime Interval

In Euclidean geometry, rotating coordinates changes x and y, but the distance $r^2 = x^2+y^2$ is invariant. In Spacetime, Lorentz transformations change x and t, but there is a quantity that remains invariant for all observers: the **Spacetime Interval** (s^2).

$$s^2 = (ct)^2 - x^2 \tag{60}$$

- If $s^2 > 0$: Time-like interval (Inside light cone). Causality holds.
- If $s^2 < 0$: Space-like interval (Outside light cone). No causality.
- If $s^2 = 0$: Light-like interval (On the cone). Connected by a light ray.

70 The Four-Vector Formalism

To handle space and time on equal footing, we introduce the concept of a Four-Vector. This notation simplifies the math of relativity and highlights the symmetries of spacetime.

70.1 Definition of the Position Four-Vector

We define a vector X with four components (X_μ, where $\mu = 0, 1, 2, 3$):

$$X = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} X_0 \\ X_1 \\ X_2 \\ X_3 \end{pmatrix} \tag{61}$$

- $X_0 = ct$: The time coordinate (scaled by c to have units of length).
- X_1, X_2, X_3: The spatial coordinates.

70.2 The Lorentz Transformation in 4D

Using $\beta = u/c$ and $\gamma = 1/\sqrt{1-\beta^2}$, the transformation for motion along x becomes symmetric:

$$X_0' = \gamma(X_0 - \beta X_1) \tag{62}$$
$$X_1' = \gamma(X_1 - \beta X_0) \tag{63}$$
$$X_2' = X_2 \tag{64}$$
$$X_3' = X_3 \tag{65}$$

This looks remarkably like a rotation, but with hyperbolic functions instead of trigonometric ones.

70.3 The Invariant Dot Product

In Euclidean space, the squared length $r^2 = x^2 + y^2$ is invariant under rotation. In Minkowski spacetime, the invariant quantity involves a minus sign. We define the relativistic dot product of two four-vectors A and B as:

$$A \cdot B = A_0 B_0 - (A_1 B_1 + A_2 B_2 + A_3 B_3) \tag{66}$$

The magnitude squared of the position vector is the spacetime interval s^2:

$$X \cdot X = c^2 t^2 - (x^2 + y^2 + z^2) = s^2$$

This value is the same for all inertial observers.

70.4 Proper Time as an Invariant

Consider a particle moving with velocity v. In a time Δt, it moves Δx. The invariant interval is:

$$\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 = c^2 \Delta t^2 (1 - v^2/c^2)$$

In the particle's own rest frame, $\Delta x' = 0$. The time measured by the particle is called **Proper Time** (τ).

$$\Delta s^2 = c^2 \Delta \tau^2$$

Equating the two expressions gives the time dilation formula:

$$\Delta \tau = \Delta t \sqrt{1 - \beta^2} = \frac{\Delta t}{\gamma} \tag{67}$$

This confirms that the time elapsed on a moving clock ($\Delta \tau$) is always less than the time measured by a stationary observer (Δt).

71 Relativistic Energy and Momentum

We seek definitions of momentum and energy that transform correctly under Lorentz transformations. To do this, we construct the Four-Momentum vector.

71.1 The Four-Velocity

Ordinary velocity $\vec{v} = d\vec{x}/dt$ is not part of a four-vector because dt changes between frames. We define the **Four-Velocity** η by differentiating with respect to proper time τ (an invariant):

$$\eta^\mu = \frac{dX^\mu}{d\tau} = \gamma \frac{dX^\mu}{dt}$$

The components are:

$$\eta = (\gamma c, \gamma v_x, \gamma v_y, \gamma v_z)$$

71.2 The Four-Momentum

We define Four-Momentum P as rest mass times Four-Velocity:

$$P^\mu = m_0 \eta^\mu = \begin{pmatrix} m_0 \gamma c \\ m_0 \gamma v_x \\ m_0 \gamma v_y \\ m_0 \gamma v_z \end{pmatrix} \tag{68}$$

The spatial part $\vec{p} = \gamma m_0 \vec{v}$ is the relativistic momentum. It grows infinitely large as $v \to c$, explaining why massive objects cannot reach light speed.

71.3 Deriving $E = mc^2$

Let's analyze the time component $P_0 = m_0 \gamma c$. Multiplying by c gives it units of energy:

$$cP_0 = m_0 c^2 \gamma = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}}$$

Using the binomial expansion for $v \ll c$:

$$cP_0 \approx m_0 c^2 \left(1 + \frac{1}{2}\frac{v^2}{c^2}\right) = m_0 c^2 + \frac{1}{2} m_0 v^2$$

We recognize $\frac{1}{2} m_0 v^2$ as the classical kinetic energy. Einstein proposed that the first term, $m_0 c^2$, is the **Rest Energy** of the particle. The total energy E is:

$$E = \gamma m_0 c^2 = m_{\text{rel}} c^2 \tag{69}$$

This reveals that mass is a form of concentrated energy. In nuclear reactions (fission/fusion), a small loss of mass (Δm) results in a huge release of energy ($\Delta E = \Delta m c^2$).

71.4 The Energy-Momentum Relation

The dot product of the Four-Momentum with itself is invariant:

$$P \cdot P = P_0^2 - \vec{p} \cdot \vec{p} = (E/c)^2 - p^2$$

Evaluating this in the rest frame (where $p = 0$ and $E = m_0 c^2$):

$$P \cdot P = (m_0 c)^2$$

Equating the general frame to the rest frame gives the fundamental relation:

$$E^2 - (pc)^2 = (m_0 c^2)^2 \tag{70}$$

This equation is vital for high-energy physics because it relates Energy, Momentum, and Mass without needing velocity. For photons ($m_0 = 0$), it yields $E = pc$.

72 Review: Constructing the Four-Vector

We seek a relativistic analog to velocity and momentum. The key is to take derivatives with respect to an **invariant** quantity.

72.1 Why define velocity with Proper Time?

Ordinary velocity $\vec{v} = d\vec{x}/dt$ is useful for calculations, but it is not a Four-Vector because dt transforms between frames. We use the invariant proper time τ (time measured by the particle itself) to define the Four-Velocity η:

$$\eta^\mu \equiv \frac{dX^\mu}{d\tau}$$

Using the time dilation relation $dt = \gamma d\tau$, we can relate this to measurable quantities:

$$\frac{dX}{d\tau} = \frac{dX}{dt}\frac{dt}{d\tau} = \gamma \frac{dX}{dt}$$

72.2 The Four-Momentum Definition

We define the Four-Momentum P as mass times Four-Velocity:

$$P^\mu = m\frac{dX^\mu}{d\tau} = (\gamma mc, \gamma mv_x, \gamma mv_y, \gamma mv_z) \tag{71}$$

This vector has two distinct parts:

1. **Spatial Part ($\vec{p}$):** The relativistic momentum.

   $$\vec{p} = \gamma m\vec{v} = \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}}$$

   At low speeds ($\gamma \approx 1$), this reduces to Newtonian momentum $\vec{p} \approx m\vec{v}$.

2. **Time Part (P_0):** Related to Energy.

   $$P_0 = \gamma mc \implies E = cP_0 = \gamma mc^2$$

   Expanding for low speeds:

   $$E \approx mc^2 + \frac{1}{2}mv^2$$

   The total energy is the sum of the **Rest Energy** (mc^2) and Kinetic Energy.

72.3 The Invariant Mass Relation

Since P is a Four-Vector, its squared length $(P \cdot P)$ must be invariant.

$$P \cdot P = P_0^2 - \vec{p} \cdot \vec{p} = (E/c)^2 - p^2$$

In the rest frame $(v = 0, p = 0, E = mc^2)$, this equals $(mc)^2$. Thus, for any observer:

$$E^2 - (pc)^2 = (mc^2)^2 \tag{72}$$

This equation is the foundation of high-energy particle physics.

72.4 Spatial vs. Time part of the Four-Momentum (short)

The four-momentum is

$$P^\mu = m\frac{dX^\mu}{d\tau} = (\gamma mc,\ \gamma m\mathbf{v}) \equiv \left(\frac{E}{c},\ \mathbf{p}\right).$$

- **Spatial part:** $\mathbf{p} = \gamma m\mathbf{v}$ is the relativistic 3-momentum. At low speeds $\mathbf{p} \approx m\mathbf{v}$.
- **Time part:** $P^0 = \gamma mc$ encodes energy via $E = cP^0 = \gamma mc^2$. This contains the rest energy mc^2 plus kinetic energy.

Physically, $\mathbf{p}$ and E are not separate unrelated quantities but components of a single four-vector because Lorentz boosts mix space and time. The invariant relation

$$P \cdot P = (E/c)^2 - \mathbf{p} \cdot \mathbf{p} = (mc)^2$$

shows how energy and momentum together determine the particle's invariant mass. Under a boost along x (speed u) the components mix:

$$E' = \gamma_u\,(E - up_x), \qquad p'_x = \gamma_u\,(p_x - uE/c^2),$$

illustrating that energy $\leftrightarrow$ momentum are frame-dependent parts of the same spacetime object.

73 Relativistic Dynamics and Collisions

The power of the four-vector formalism becomes clear when analyzing collisions. The total four-momentum is always conserved.

$$P^\mu_{\text{total, initial}} = P^\mu_{\text{total, final}}$$

This single vector equation represents both Conservation of Energy (time component) and Conservation of Momentum (space components).

73.1 Mass-Energy Equivalence in Collisions

In Newtonian mechanics, kinetic energy is often lost in inelastic collisions. In relativity, total energy E is always conserved. "Lost" kinetic energy is converted into mass.

Example: Creating Mass Two particles of mass m travel towards each other with speed v and stick together to form a single particle of mass M.

- **Initial State:** Particle 1: $P_1 = (\gamma mc, \gamma mv)$
  Particle 2: $P_2 = (\gamma mc, -\gamma mv)$
- **Final State:** Particle M is at rest (by symmetry). $P_M = (Mc, 0)$

Conservation of Four-Momentum ($P_1 + P_2 = P_M$):

$$\begin{pmatrix} \gamma mc + \gamma mc \\ \gamma mv - \gamma mv \end{pmatrix} = \begin{pmatrix} Mc \\ 0 \end{pmatrix} \implies 2\gamma mc = Mc$$

Thus, the final mass is:

$$M = 2\gamma m = \frac{2m}{\sqrt{1 - v^2/c^2}} \tag{73}$$

Since $\gamma > 1$, $M > 2m$. The kinetic energy of the incoming particles has been converted into the rest mass of the final product.

73.2 Photons: Massless Particles

A photon has mass $m = 0$ but carries energy and momentum. From the invariant mass relation $E^2 - p^2c^2 = m^2c^4$:

$$m = 0 \implies E = pc$$

The four-momentum of a photon traveling in the x-direction is:

$$K^\mu = \begin{pmatrix} E/c \\ p \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} p \\ p \\ 0 \\ 0 \end{pmatrix} \tag{74}$$

The magnitude squared of a photon's four-momentum is always zero:

$$K \cdot K = p^2 - p^2 = 0$$

Such vectors are called **null vectors** or light-like vectors.

73.3 Absorption of a Photon

Consider a stationary atom of mass M absorbing a photon of energy E_γ.

- Initial Four-Momentum: $P_{\text{total}} = P_{\text{atom}} + P_\gamma = (Mc, 0) + (E_\gamma/c, E_\gamma/c)$.
- Final Four-Momentum: $P_{\text{final}} = (E_{\text{new}}/c, p_{\text{new}})$.

Conservation implies:

$$E_{\text{new}} = Mc^2 + E_\gamma, \quad p_{\text{new}} = E_\gamma/c$$

The new mass M' is found using the invariant dot product $P_{\text{final}} \cdot P_{\text{final}} = (M'c)^2$:

$$(P_{\text{atom}} + Pkheatombecomesheavier($$

M' ¿ M)$andrecoils.$

74 Particle Physics Application: Threshold Energy

A common problem in high-energy physics is calculating the minimum energy required to create new particles.

74.1 The Problem

We want to create a proton-antiproton pair by firing a proton at a stationary proton target.

$$p + p \to p + p + p + \bar{p} \tag{75}$$

- Mass of proton $m_p = m$.
- Initial State: A moving proton $(E, \vec{p})$ and a target proton $(mc^2, 0)$.
- Final State: Four particles of mass m.

What is the minimum Energy E required?

74.2 The Method of Invariants

Instead of solving for velocities, we use the invariance of the squared total four-momentum $(P_{\text{tot}} \cdot P_{\text{tot}})$.

1. Evaluate in the Center of Mass (CM) Frame
In the CM frame, the total momentum is zero. The minimum energy scenario is when all product particles are created **at rest**.

$$P_{\text{tot, CM}} = (4mc, 0, 0, 0)$$

The invariant square is:

$$P \cdot P = (4mc)^2 - 0 = 16m^2c^2$$

2. Evaluate in the Lab Frame
In the lab, the total four-momentum is the sum of the beam and the target:

$$P_{\text{tot, Lab}} = P_{\text{beam}} + P_{\text{target}} = \left(\frac{E}{c} + mc, \vec{p}\right)$$

Squaring this:

$$P \cdot P = \left(\frac{E}{c} + mc\right)^2 - p^2$$

$$P \cdot P = \frac{E^2}{c^2} + m^2c^2 + 2Em - p^2$$

Recall that for the beam proton, $E^2/c^2 - p^2 = m^2c^2$. Substituting this:

$$P \cdot P = (m^2c^2) + m^2c^2 + 2Em = 2m^2c^2 + 2Em$$

3. Equate and Solve
Since $P \cdot P$ is invariant:

$$\begin{aligned} 16m^2c^2 &= 2m^2c^2 + 2Em \\ 14m^2c^2 &= 2Em \\ E &= 7mc^2 \end{aligned} \tag{76}$$

Conclusion: To create $2m$ of new mass, we need an incoming energy of $7mc^2$.

- Energy used to create mass: $2mc^2$.
- Energy used to conserve momentum (kinetic): $4mc^2$.
- Initial energy of beam proton: mc^2 (rest) + $6mc^2$ (kinetic).

This inefficiency (needing $7mc^2$ to do $4mc^2$ of work) drives the design of **Colliders**, where two beams hit head-on. In a collider, the Lab Frame is the Center of Mass Frame, so all energy is available for particle creation.

75 Mathematical Methods: Taylor Series and Complex Numbers

To solve physics problems involving oscillations and waves, we need powerful mathematical tools to approximate functions and handle periodic motion.

75.1 The Taylor Series

Any smooth function $f(x)$ can be approximated near a point (say $x = 0$) by a polynomial. We construct this polynomial so that its value, slope, curvature, and higher derivatives match the original function at $x = 0$.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \dots \tag{77}$$

The $n!$ (factorial) term compensates for the repeated differentiation of x^n.

75.2 Important Series Expansions

- **The Exponential Function:** (Valid for all x)

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

- **Cosine:** (Even function, even powers)

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

- **Sine:** (Odd function, odd powers)

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

75.3 Euler's Formula

What happens if we raise e to an imaginary power ix, where $i = \sqrt{-1}$? We use the power series definition:

$$e^{ix} = 1 + (ix) + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \dots$$

Recall the powers of i: $i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1$. Grouping real and imaginary terms:

$$e^{ix} = \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + i\left(x - \frac{x^3}{3!} + \dots\right)$$

We recognize these series as cosine and sine. Thus, we arrive at **Euler's Formula**:

$$e^{ix} = \cos x + i \sin x \tag{78}$$

Key Identities:

- $\cos x = \frac{e^{ix} + e^{-ix}}{2}$

- $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$
- $e^{i\pi} + 1 = 0$ (Euler's Identity)

This connects exponential growth to oscillatory motion, which is fundamental for solving the harmonic oscillator.

76 Applications of Euler's Formula and Taylor Series

The mathematical tools developed above (Complex Numbers and Series) are essential for solving the physics of oscillations.

76.1 Complex Numbers in Physics

A complex number $z = x + iy$ can be written in polar form using Euler's formula:

$$z = re^{i\theta} = r(\cos\theta + i\sin\theta)$$

where $r = \sqrt{x^2 + y^2}$ is the magnitude and θ is the phase. **Why do we use this?** Multiplying exponentials is easier than multiplying trigonometric functions.

$$z_1 z_2 = (r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

Multiplication corresponds to scaling the length and rotating the angle. This geometry underlies the study of waves and AC circuits.

76.2 Simple Harmonic Motion (SHM)

Consider a mass m on a spring with constant k. The force is $F = -kx$. Newton's Law:

$$m\frac{d^2x}{dt^2} = -kx \quad \Rightarrow \quad \frac{d^2x}{dt^2} = -\omega^2 x$$

where $\omega = \sqrt{k/m}$. Using our knowledge of derivatives, we know functions that return themselves with a minus sign after two derivatives are sines and cosines (or complex exponentials). The general solution is:

$$x(t) = A\cos(\omega t + \phi) \tag{79}$$

- A: Amplitude (determined by initial pull).
- ϕ: Phase (determined by when you start the clock).
- ω: Frequency (determined by the system k, m, NOT by initial conditions).

76.3 The Simple Pendulum

A mass m hangs from a massless rod of length L. Gravity exerts a torque $\tau = -mgL\sin\theta$. Newton's Rotational Law ($\tau = I\alpha$):

$$mL^2\frac{d^2\theta}{dt^2} = -mgL\sin\theta$$

This differential equation is non-linear and difficult to solve exactly. However, for small angles, we use the **Taylor Series approximation**:

$$\sin\theta \approx \theta - \frac{\theta^3}{6} + \cdots \approx \theta$$

Substituting this linearizes the equation:

$$mL^2\frac{d^2\theta}{dt^2} \approx -mgL\theta \implies \frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$$

This is exactly the SHM equation with $\omega = \sqrt{g/L}$.

$$T = 2\pi\sqrt{\frac{L}{g}} \tag{80}$$

This derivation shows that the pendulum's period is independent of mass and amplitude, a result only valid within the small-angle approximation.

77 Rotational Oscillations

Simple Harmonic Motion applies to any system where the restoring force (or torque) is proportional to the displacement.

77.1 Torsional Pendulum

Consider a disk suspended by a wire. If we twist the wire by an angle θ, it exerts a restoring torque:

$$\tau = -\kappa\theta$$

where κ is the torsion constant. Newton's Rotational Law ($\tau = I\alpha$) gives:

$$I\frac{d^2\theta}{dt^2} = -\kappa\theta \implies \frac{d^2\theta}{dt^2} = -\left(\frac{\kappa}{I}\right)\theta$$

By analogy with the spring ($k \to \kappa, m \to I$), the angular frequency is:

$$\omega = \sqrt{\frac{\kappa}{I}} \tag{81}$$

77.2 The Physical Pendulum

A "Physical Pendulum" is any rigid body of mass M suspended from a pivot point P that is not its center of mass. Let d be the distance from the pivot P to the Center of Mass (CM).

- **Torque:** Gravity acts at the CM. The torque about P is:

$$\tau = -Mgd\sin\theta \approx -(Mgd)\theta \quad \text{(for small } \theta\text{)}$$

- **Effective Constant:** Comparing to $\tau = -\kappa\theta$, we see $\kappa = Mgd$.
- **Moment of Inertia:** I must be calculated about the pivot P. Using the Parallel Axis Theorem:

$$I_P = I_{\text{CM}} + Md^2$$

The frequency of oscillation is:

$$\omega = \sqrt{\frac{Mgd}{I_P}} \implies T = 2\pi\sqrt{\frac{I_P}{Mgd}} \tag{82}$$

Check: For a simple pendulum (point mass m at distance L), $I_P = mL^2$ and $d = L$.

$$\omega = \sqrt{\frac{mgL}{mL^2}} = \sqrt{\frac{g}{L}}$$

This matches our previous result.

77.3 General Strategy for Finding Frequency

To find the frequency of any oscillating system:

1. Find the equilibrium position.
2. Displace the system by a small amount (x or θ).
3. Calculate the restoring force F or torque τ.
4. Show that for small displacements, $F \approx -k_{\text{eff}}x$ or $\tau \approx -\kappa_{\text{eff}}\theta$.
5. The frequency is $\omega = \sqrt{k_{\text{eff}}/m}$ or $\omega = \sqrt{\kappa_{\text{eff}}/I}$.

78 Solving Oscillations with Complex Numbers

While sines and cosines work for simple springs, complex exponentials provide a much more powerful method for solving differential equations, especially when friction (damping) is involved.

78.1 The Complex Ansatz

We want to solve the SHM equation:

$$\ddot{x} + \omega_0^2 x = 0$$

We guess a solution of the form $x(t) = Ae^{\alpha t}$. Substituting this into the equation:

$$\alpha^2 Ae^{\alpha t} + \omega_0^2 Ae^{\alpha t} = 0$$

Dividing by $Ae^{\alpha t}$ (which is never zero):

$$\alpha^2 + \omega_0^2 = 0 \implies \alpha = \pm i\omega_0$$

Thus, we find two independent solutions: $e^{i\omega_0 t}$ and $e^{-i\omega_0 t}$.

78.2 Linearity and Superposition

The harmonic oscillator equation is **Linear**.

Definition: An equation is linear if the dependent variable x and its derivatives appear only to the first power (no x^2, $\sin(x)$, etc.).

Principle of Superposition: If $x_1(t)$ and $x_2(t)$ are solutions to a linear homogeneous equation, then any linear combination is also a solution:

$$x(t) = C_1 x_1(t) + C_2 x_2(t) \tag{83}$$

Using our two exponential solutions, the general solution is:

$$x(t) = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}$$

where A and B are arbitrary complex constants.

78.3 Recovering the Real Solution

Physical position $x(t)$ must be a real number. A number is real if it equals its complex conjugate ($z = z^*$).

$$x^*(t) = A^* e^{-i\omega_0 t} + B^* e^{i\omega_0 t}$$

Setting $x(t) = x^*(t)$ implies that we must choose $B = A^*$. Let $A = \frac{C}{2}e^{i\phi}$ (polar form). Then $B = \frac{C}{2}e^{-i\phi}$.

$$x(t) = \frac{C}{2}e^{i(\omega_0 t + \phi)} + \frac{C}{2}e^{-i(\omega_0 t + \phi)}$$

Using Euler's identity ($e^{iu} + e^{-iu} = 2\cos u$):

$$x(t) = C\cos(\omega_0 t + \phi) \tag{84}$$

We have recovered the familiar cosine solution, but the complex exponential method will prove superior for damped and driven oscillators.

79 Damped and Driven Oscillations

Real-world oscillators experience friction (damping) and are often subject to external driving forces. The complex exponential method simplifies the analysis significantly.

79.1 Damped Harmonic Motion

Consider a mass on a spring with a damping force proportional to velocity ($F_{\text{drag}} = -b\dot{x}$). The equation of motion is:

$$m\ddot{x} + b\dot{x} + kx = 0 \quad \Rightarrow \quad \ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

where $\gamma = b/m$ and $\omega_0 = \sqrt{k/m}$. Using the ansatz $x(t) = e^{\alpha t}$, the characteristic equation is:

$$\alpha^2 + \gamma\alpha + \omega_0^2 = 0$$

The roots are:

$$\alpha = -\frac{\gamma}{2} \pm \sqrt{\left(\frac{\gamma}{2}\right)^2 - \omega_0^2}$$

For weak damping (Underdamped, $\gamma/2 < \omega_0$), the term under the square root is negative, leading to complex roots:

$$\alpha = -\frac{\gamma}{2} \pm i\omega' \quad \text{where } \omega' = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}}$$

The physical solution is:

$$x(t) = Ae^{-\frac{\gamma}{2}t}\cos(\omega' t + \phi) \tag{85}$$

Physical Meaning: The system oscillates at a slightly reduced frequency ω', while the amplitude decays exponentially ($e^{-\frac{\gamma}{2}t}$) due to energy loss.

79.2 Driven Harmonic Motion

Now apply an external driving force $F(t) = F_0\cos(\omega t)$.

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos(\omega t)$$

The Complex Trick: Instead of solving this directly, we solve a complex equation with driving force $F_0 e^{i\omega t}$ and take the Real part at the end. Let $z(t) = z_0 e^{i\omega t}$. Substituting into the equation:

$$(-\omega^2 + i\omega\gamma + \omega_0^2)z_0 e^{i\omega t} = \frac{F_0}{m}e^{i\omega t}$$

Solving for the complex amplitude z_0:

$$z_0 = \frac{F_0/m}{(\omega_0^2 - \omega^2) + i(\gamma\omega)}$$

To find the physical amplitude and phase, we write the denominator in polar form $Re^{i\phi}$:

$$R = \sqrt{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2} \quad \text{and} \quad \tan\phi = \frac{\gamma\omega}{\omega_0^2 - \omega^2}$$

The physical solution $x(t) = \mathrm{Re}(z(t))$ is:

$$x(t) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}} \cos(\omega t - \phi) \tag{86}$$

79.3 Resonance

The amplitude of oscillation depends heavily on the driving frequency ω.

- **Low Frequency ($\omega \to 0$):** Amplitude $\approx \frac{F_0}{m\omega_0^2} = \frac{F_0}{k}$. (Spring stiffness dominates).
- **High Frequency ($\omega \to \infty$):** Amplitude $\to 0$. (Inertia dominates).
- **Resonance ($\omega \approx \omega_0$):** The term $(\omega_0^2 - \omega^2)$ vanishes. The denominator is minimized, and amplitude becomes maximum:

$$A_{\max} \approx \frac{F_0}{m\gamma\omega_0}$$

At resonance, the driving force is perfectly synchronized with the natural motion, pumping energy into the system efficiently. The only limit to the amplitude is the damping γ.

80 Damped and Driven Harmonic Motion

80.1 Definitions

The equation of motion for a mass m on a spring k with damping b and driving force $F_0 \cos(\omega t)$ is:

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos(\omega t) \tag{87}$$

- $\omega_0 = \sqrt{k/m}$: **Natural Frequency** (The system's intrinsic vibration speed).
- $\gamma = b/m$: **Damping Coefficient** (Measure of friction).
- ω: **Driving Frequency** (Frequency of the external force).

80.2 Free Vibration (No Driving Force)

When $F_0 = 0$, the system behaves according to the level of damping:

1. **Underdamped** ($\gamma < 2\omega_0$): The system oscillates with decaying amplitude.

$$x(t) = Ae^{-\frac{\gamma}{2}t}\cos(\omega' t + \phi)$$

2. **Overdamped** ($\gamma > 2\omega_0$): The system returns to equilibrium without oscillating.
3. **Critically Damped** ($\gamma = 2\omega_0$): The system returns to equilibrium as fast as possible. (Used in shock absorbers).

80.3 Driven Vibration (Resonance)

When driven by an external force, we use complex exponentials to solve the equation. The steady-state solution is:

$$x(t) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}}\cos(\omega t - \phi) \tag{88}$$

Resonance: When the driving frequency ω is close to the natural frequency ω_0, the denominator becomes very small (impedance is minimized), resulting in a massive increase in Amplitude.

81 The General Solution for Driven Oscillators

The differential equation for a driven, damped oscillator is linear. This means the general solution $x(t)$ is the sum of the **Particular Solution** (steady state) and the **Complementary Solution** (transient).

$$x(t) = x_p(t) + x_c(t) \tag{89}$$

81.1 The Particular Solution (x_p)

This describes the motion after the system has settled down. It oscillates at the driving frequency ω with a fixed amplitude determined by the force and impedance.

$$x_p(t) = A(\omega)\cos(\omega t - \phi)$$

where $A(\omega) = \frac{F_0/m}{|Z|}$. Note that x_p has **no arbitrary constants**. It is completely determined by the driving force.

81.2 The Complementary Solution (x_c)

This is the solution to the homogeneous equation (with $F = 0$). It represents the natural, decaying vibration of the system.

$$x_c(t) = Ce^{-\frac{\gamma}{2}t}\cos(\omega' t + \phi_0)$$

This part contains two arbitrary constants (C and ϕ_0).

81.3 Matching Initial Conditions

Why do we need x_c if it dies out? Because at $t = 0$, the mass has a specific position $x(0)$ and velocity $v(0)$. The steady state solution x_p usually does not match these values. We use the constants C and ϕ_0 in the transient term to satisfy:

$$x(0) = x_p(0) + x_c(0)$$
$$v(0) = \dot{x}_p(0) + \dot{x}_c(0)$$

Physical Interpretation: When you first turn on a machine, it shakes transiently (a mix of natural and driving frequencies) before settling into a smooth steady hum. The transient is the x_c term dying out.

82 Introduction to Waves

A wave is a disturbance that propagates through a medium. Unlike a single oscillator (one coordinate $x(t)$), a wave involves a continuous medium where every point x has a displacement $\psi(x, t)$.

82.1 Types of Waves

- **Transverse Waves:** The displacement is perpendicular to the motion (e.g., a wave on a string).
- **Longitudinal Waves:** The displacement is parallel to the motion (e.g., sound waves).

82.2 Derivation of the Wave Equation (String)

Consider a small segment of a string with mass density μ (kg/m) under tension T. We apply Newton's Second Law to a segment of length dx.

- **Mass:** $dm = \mu dx$.

- **Acceleration:** $a_y = \frac{\partial^2 \psi}{\partial t^2}$.
- **Net Force:** The vertical component of tension acts on both ends at slightly different angles θ.

$$F_y \approx T \frac{\partial \psi}{\partial x}\bigg|_{x+dx} - T \frac{\partial \psi}{\partial x}\bigg|_{x} \approx T \frac{\partial^2 \psi}{\partial x^2} dx$$

Equating $F = ma$:

$$T \frac{\partial^2 \psi}{\partial x^2} dx = (\mu dx) \frac{\partial^2 \psi}{\partial t^2}$$

Dividing by dx and rearranging gives the **1D Wave Equation**:

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \tag{90}$$

where $v = \sqrt{T/\mu}$ is the wave propagation speed.

82.3 The Traveling Wave Solution

The general solution to the wave equation is any function of the form $f(x \pm vt)$. A specific sinusoidal solution traveling to the right is:

$$\psi(x, t) = A \cos(kx - \omega t) \tag{91}$$

Parameters:

- A: Amplitude (maximum displacement).
- k: Wavenumber ($k = 2\pi/\lambda$).
- ω: Angular Frequency ($\omega = 2\pi f$).
- v: Wave Speed ($v = \omega/k = \lambda f$).

Why $x - vt$? Consider a pulse $f(x)$ centered at $x = 0$. To shift the graph to the right by a distance d, we write $f(x - d)$. If the pulse moves at velocity v, the distance is $d = vt$, so the moving pulse is described by $f(x - vt)$.

- $f(x - vt)$: Wave moving Right.
- $f(x + vt)$: Wave moving Left.

83 Anatomy of a Sinusoidal Wave

While any function $f(x - vt)$ is a wave, the most common and useful wave form is the sinusoidal wave (also called a harmonic wave or plane wave).

$$\psi(x, t) = A \cos(kx - \omega t) \tag{92}$$

83.1 Spatial Parameters (t fixed)

Take a snapshot of the wave at $t = 0$. The shape is $\psi(x) = A\cos(kx)$. The wave repeats in space after a distance called the **Wavelength** (λ). Since the cosine function repeats every 2π radians:

$$k\lambda = 2\pi \implies k = \frac{2\pi}{\lambda}$$

k is called the **Wavenumber** (units: rad/m). It represents spatial frequency.

83.2 Temporal Parameters (x fixed)

Observe a single point on the string at $x = 0$. The motion is $\psi(t) = A\cos(-\omega t) = A\cos(\omega t)$. The motion repeats in time after a duration called the **Period** (T).

$$\omega T = 2\pi \implies \omega = \frac{2\pi}{T} = 2\pi f$$

ω is the **Angular Frequency** (units: rad/s). f is the standard frequency (Hz).

83.3 Phase Velocity

The speed of the wave v relates the spatial and temporal properties. From the wave equation, we found $v = \omega/k$. Substituting the definitions above:

$$v = \frac{2\pi/T}{2\pi/\lambda} = \frac{\lambda}{T}$$

Since $f = 1/T$, we arrive at the fundamental wave relationship:

$$v = \lambda f \tag{93}$$

Physical Interpretation: If a source emits f waves per second, and each wave is λ meters long, the wave train advances by λf meters every second.

83.4 Summary of Relationships

Variable	Symbol	Relation
Wavelength	λ	$2\pi/k$
Wavenumber	k	$2\pi/\lambda$
Period	T	$2\pi/\omega$
Frequency	f	$1/T = \omega/2\pi$
Angular Freq	ω	$2\pi f$
Velocity	v	$\lambda f = \omega/k$

84 Energy, Power, and Intensity of Waves

Waves transport energy through a medium. We calculate this energy by treating the medium (string) as a collection of harmonic oscillators.

84.1 Energy Density

Consider a small segment of string of length dx and mass $dm = \mu dx$. When the segment passes through equilibrium, its energy is entirely kinetic.

$$v_{\text{max}} = \omega A$$

The energy in this segment is:

$$dE = \frac{1}{2} dm v_{\text{max}}^2 = \frac{1}{2}(\mu dx)(\omega A)^2$$

The **Energy Density** (Energy per unit length) is:

$$\frac{dE}{dx} = \frac{1}{2}\mu\omega^2 A^2 \tag{94}$$

This confirms that wave energy is proportional to the square of the amplitude (A^2) and the square of the frequency (ω^2).

84.2 Power Transmitted

Power is the rate at which energy is delivered: $P = dE/dt$. In time dt, the wave travels a distance $dx = vdt$. The energy contained in this length is delivered to the next section.

$$P = \frac{dE}{dx} \cdot \frac{dx}{dt} = \left(\frac{1}{2}\mu\omega^2 A^2\right) v$$

$$P = \frac{1}{2}\mu v \omega^2 A^2 \tag{95}$$

84.3 Sound Intensity and Decibels

For 3D waves (like sound), we define **Intensity** (I) as Power per unit Area. For a point source emitting power P, the intensity at distance r is:

$$I = \frac{P}{4\pi r^2}$$

Because the human ear detects a vast range of intensities, we use a logarithmic scale called the **Decibel (dB)**:

$$\beta = 10 \log_{10}\left(\frac{I}{I_0}\right) \tag{96}$$

where $I_0 = 10^{-12}\,\text{W/m}^2$ is the threshold of human hearing.

85 The Doppler Effect

The observed frequency of a wave changes if the source or the observer is moving relative to the medium.

85.1 Case 1: Moving Source (The "Squashed Wavelength")

A source emits sound of frequency f and speed v, while moving towards a stationary observer at speed u_s.

- In one period T, the wave travels $vT = \lambda$.
- The source also travels $u_s T$.
- The effective wavelength is "squashed": $\lambda' = vT - u_s T$.

The observed frequency f' is:

$$f' = \frac{v}{\lambda'} = \frac{v}{(v - u_s)T} = \frac{v}{v - u_s} f$$

$$f' = f\left(\frac{1}{1 - u_s/v}\right) \tag{97}$$

The correction is in the **Denominator**. As $u_s \to v$ (sound barrier), the frequency goes to infinity (Sonic Boom).

85.2 Case 2: Moving Observer (The "collision Rate")

An observer moves towards a stationary source at speed u_o.

- The wavelength λ in the air is unchanged.
- The relative speed of the waves hitting the observer is $v' = v + u_o$.

The observed frequency is:

$$f' = \frac{v'}{\lambda} = \frac{v + u_o}{v/f} = f\frac{v + u_o}{v}$$

$$f' = f\left(1 + \frac{u_o}{v}\right) \tag{98}$$

The correction is in the **Numerator**.

85.3 Summary of Doppler Signs

$$f' = f\left(\frac{v \pm u_o}{v \mp u_s}\right)$$

- **Top Sign (+/-):** Observer moving Towards/Away.
- **Bottom Sign (-/+):** Source moving Towards/Away.

86 Interference and Superposition

The wave equation is linear, meaning that if ψ_1 and ψ_2 are solutions, then $\psi_{\text{total}} = \psi_1 + \psi_2$ is also a solution. When waves overlap, they interfere.

86.1 Beats (Interference in Time)

Consider two sound waves of equal amplitude but slightly different frequencies ω_1 and ω_2 reaching an observer at $x = 0$.

$$\psi(t) = A\cos(\omega_1 t) + A\cos(\omega_2 t)$$

Using the trigonometric identity $\cos A + \cos B = 2\cos\frac{A-B}{2}\cos\frac{A+B}{2}$:

$$\psi(t) = \left[2A\cos\left(\frac{\omega_1 - \omega_2}{2}t\right)\right]\cos\left(\frac{\omega_1 + \omega_2}{2}t\right)$$

- **Carrier:** The fast oscillation at the average frequency.
- **Envelope:** The slow amplitude modulation (beats).

The intensity varies with the beat frequency:

$$f_{\text{beat}} = |f_1 - f_2| \tag{99}$$

86.2 Standing Waves (Interference in Space)

When a wave reflects off a fixed boundary, it inverts. The superposition of the incident wave (traveling right) and the reflected wave (traveling left) creates a Standing Wave.

$$\psi = A\sin(kx - \omega t) + A\sin(kx + \omega t) = 2A\sin(kx)\cos(\omega t)$$

This wave does not travel. The shape $\sin(kx)$ just oscillates up and down.

- **Nodes:** Points where $\sin(kx) = 0$. The string never moves here.
- **Antinodes:** Points where amplitude is maximum ($2A$).

86.3 Modes of Vibration

For a string of length L fixed at both ends, the boundary conditions require nodes at $x = 0$ and $x = L$. This restricts the possible wavelengths:

$$\sin(kL) = 0 \implies kL = n\pi \implies \frac{2\pi}{\lambda}L = n\pi$$

$$\lambda_n = \frac{2L}{n}, \quad f_n = n\left(\frac{v}{2L}\right) = nf_1 \tag{100}$$

where $n = 1, 2, 3 \ldots$. These are the **Harmonics** of the string.

86.4 Sound in Pipes

- **Open-Open Pipe (Flute):** Ends are antinodes. Behaves like a string. All harmonics ($f_1, 2f_1, 3f_1 \ldots$) are present.
- **Closed-Open Pipe (Bottle):** One node (closed end) and one antinode (open end). This fits only odd quarters of a wavelength ($\lambda/4, 3\lambda/4 \ldots$).

  $$f_n = n\left(\frac{v}{4L}\right) \quad \text{for } n = 1, 3, 5 \ldots$$

  Only **odd harmonics** are present.

87 Fluid Statics

Fluids (liquids and gases) are substances that cannot resist shear stress when at rest. They conform to the shape of their container.

87.1 Density and Pressure

- **Density (ρ):** Mass per unit volume.

  $$\rho = \frac{m}{V} \quad (\text{Water: } 1000 \text{ kg/m}^3)$$

- **Pressure (P):** The normal force exerted by a fluid per unit area.

  $$P = \frac{F}{A} \tag{101}$$

  Units: Pascal (Pa) = N/m^2. Standard Atmospheric Pressure: $P_{\text{atm}} \approx 1.01 \times 10^5$ Pa.

87.2 Variation of Pressure with Depth

Consider a static fluid column of height h and cross-sectional area A. The fluid at the bottom must support the weight of the fluid above it plus the pressure at the top.

- Downward Force: $F_{\text{top}} + W = P_0 A + (\rho A h)g$.
- Upward Force: $F_{\text{bottom}} = PA$.

For equilibrium, forces must balance:

$$PA = P_0 A + \rho g h A$$

Dividing by A:

$$P = P_0 + \rho g h \tag{102}$$

This is the **Hydrostatic Pressure Law**. Pressure increases linearly with depth.

87.3 Pascal's Principle and Hydraulics

Pascal's Principle states that a change in pressure applied to an enclosed fluid is transmitted undiminished to every point of the fluid and the walls of the container.

Application: The Hydraulic Lift
Consider two pistons connected by a fluid.

- Input: Force F_1 on small area A_1. Pressure increase $\Delta P = F_1/A_1$.
- Output: The same pressure ΔP acts on larger area A_2.

$$F_2 = \Delta P \cdot A_2 = \frac{F_1}{A_1} A_2$$

$$F_2 = F_1 \left(\frac{A_2}{A_1} \right) \tag{103}$$

Since $A_2 > A_1$, the output force is magnified. This allows a small force to lift a heavy car. However, energy is conserved. The work done is the same:

$$F_1 d_1 = F_2 d_2$$

To lift the heavy load a small distance d_2, the input piston must move a large distance d_1.

88 Buoyancy and Fluid Dynamics

88.1 Archimedes' Principle

Any object submerged in a fluid experiences an upward buoyant force equal to the weight of the fluid displaced by the object.

$$F_B = \rho_{\text{fluid}} V_{\text{submerged}} g \tag{104}$$

Applications:

- **Sinking:** If $\rho_{\text{obj}} > \rho_{\text{fluid}}$, the weight ($mg$) exceeds F_B. The apparent weight is $W' = mg - F_B$.
- **Floating:** If $\rho_{\text{obj}} < \rho_{\text{fluid}}$, the object floats. It sinks only deep enough to displace its own weight in fluid.

$$\frac{V_{\text{submerged}}}{V_{\text{total}}} = \frac{\rho_{\text{obj}}}{\rho_{\text{fluid}}}$$

89 Bernoulli's Principle

When a fluid moves, we apply the Conservation of Energy to a flowing volume element. We assume the fluid is incompressible and non-viscous (ideal).

89.1 The Equation of Continuity

Since mass is conserved, the volume flow rate (Q) must be constant throughout a pipe, even if the area changes.

$$A_1 v_1 = A_2 v_2$$

This means fluid speeds up in narrow constrictions ($A \downarrow \implies v \uparrow$).

89.2 Bernoulli's Equation

Applying the Work-Energy theorem to a fluid element:

$$\text{Work by Pressure} = \Delta K + \Delta U$$

Rearranging terms yields the conservation law along a streamline:

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{Constant} \tag{105}$$

This famous equation links pressure, speed, and height.

89.3 Key Implication: The Venturi Effect

If a fluid flows horizontally ($h_1 = h_2$) through a constriction:

- The speed increases (Continuity).
- Therefore, the pressure **decreases** (Bernoulli).

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

Since $v_2 > v_1$, it must be that $P_2 < P_1$. This principle explains **lift on airplane wings** (faster air on top = lower pressure = upward force) and **Torricelli's Law** (speed of efflux from a tank $v = \sqrt{2gh}$).

90 Introduction to Thermodynamics

Thermodynamics is the study of heat, work, temperature, and energy on a macroscopic scale. We begin by defining the fundamental concept of temperature.

90.1 Thermal Equilibrium and the Zeroth Law

When two objects at different temperatures are placed in contact, they exchange energy until their macroscopic properties (like pressure and volume) stop changing. This state is called **Thermal Equilibrium**.

The Zeroth Law of Thermodynamics: If system A is in thermal equilibrium with system B, and system B is in thermal equilibrium with system C, then system A is in thermal equilibrium with system C.

- This law justifies the use of **thermometers**. If B is a thermometer, it tells us that A and C have the same "Temperature."

90.2 Temperature Scales

To measure temperature, we need a property that changes with it (thermometric property).

1. **Liquid Expansion:** Mercury or alcohol in a glass tube. We define two fixed points:
   - Freezing point of water = 0°C.
   - Boiling point of water = 100°C.

2. **Gas Thermometers:** For dilute gases at constant volume, Pressure is linear with Temperature.

$$P \propto T$$

Gas thermometers are preferred because all dilute gases behave identically, providing a universal standard.

90.3 Absolute Zero and the Kelvin Scale

If we plot Pressure vs. Temperature for any gas, the lines all extrapolate to zero pressure at a specific temperature: $-273.15°$C. This defines the absolute lower limit of temperature, **Absolute Zero**. We define the **Kelvin Scale** (Absolute Temperature) by shifting the zero point:

$$T_{\text{Kelvin}} = T_{\text{Celsius}} + 273.15 \tag{106}$$

- Absolute Zero: 0 K.
- Triple Point of Water (where ice, water, and steam coexist): Defined exactly as 273.16 K.

90.4 Heat (Q)

Heat is the energy transferred between a system and its environment due to a temperature difference.

- **Historical Unit:** The **calorie** (cal) is the heat required to raise the temperature of 1 gram of water by 1°C.
- **Modern Unit:** Since heat is energy, the SI unit is the **Joule** (J).
- **Conversion:** 1 cal = 4.184 J.

91 Specific Heat and Calorimetry

How much heat is required to change the temperature of an object? This depends on the amount of substance, the desired temperature change, and the nature of the material itself.

91.1 Specific Heat Capacity (c)

The heat Q required to change the temperature of a mass m by an amount ΔT is given by:

$$Q = mc\Delta T \tag{107}$$

where c is the **Specific Heat Capacity** of the material.

- **Water:** $c \approx 1$ cal/g·°C (or 4184 J/kg·K). Water has a very high specific heat, meaning it resists temperature changes (which is why oceans stabilize the climate).
- **Metals:** Generally have low specific heat (e.g., Copper ≈ 0.09 cal/g·°C), meaning they heat up and cool down quickly.

91.2 Thermal Expansion

Similarly, materials expand when heated. For a solid rod of length L, the change in length is proportional to the temperature change:

$$\Delta L = \alpha L \Delta T \tag{108}$$

where α is the **Coefficient of Linear Expansion**. For volume expansion (liquids or solids):

$$\Delta V = \beta V \Delta T$$

where $\beta \approx 3\alpha$ is the coefficient of volume expansion.

91.3 Calorimetry: The Method of Mixtures

When two objects at different temperatures are placed in thermal contact and isolated from the rest of the universe, they exchange heat until they reach a common final temperature T_f. By the Law of Conservation of Energy:

$$\text{Heat Lost by Hot Body} + \text{Heat Gained by Cold Body} = 0$$

$$\sum Q_i = 0 \quad \Rightarrow \quad \sum m_i c_i (T_f - T_{i,\text{initial}}) = 0 \tag{109}$$

Example: Dropping mass m_1 (specific heat c_1, temp T_1) into water of mass m_2 (specific heat c_2, temp T_2).

$$m_1 c_1 (T_f - T_1) + m_2 c_2 (T_f - T_2) = 0$$

Solving for T_f gives the final equilibrium temperature.

92 Phase Changes and Heat Transfer

When heat is added to a substance, it usually increases in temperature. However, at specific transition points (melting and boiling), the temperature remains constant while the substance changes phase.

92.1 Latent Heat (L)

The energy required to change the phase of a mass m without changing its temperature is:

$$Q = mL \tag{110}$$

where L is the **Latent Heat**.

- **Latent Heat of Fusion (L_f):** Solid $\leftrightarrow$ Liquid. (Water: $L_f \approx 80$ cal/g).
- **Latent Heat of Vaporization (L_v):** Liquid $\leftrightarrow$ Gas. (Water: $L_v \approx 540$ cal/g).

The Heating Curve:

1. Heat ice from $-T$ to 0°C ($Q = mc_{\text{ice}}\Delta T$).
2. Melt ice at 0°C ($Q = mL_f$). Temp is constant.
3. Heat water from 0°C to 100°C ($Q = mc_{\text{water}}\Delta T$).
4. Boil water at 100°C ($Q = mL_v$). Temp is constant.
5. Heat steam above 100°C ($Q = mc_{\text{steam}}\Delta T$).

92.2 Mechanisms of Heat Transfer

- **Conduction:** Direct transfer of vibrational energy through a solid material. The rate of heat flow ($H = dQ/dt$) is proportional to the cross-sectional area A and the temperature gradient:

$$H = -\kappa A \frac{dT}{dx} \approx \kappa A \frac{T_{\text{hot}} - T_{\text{cold}}}{L} \tag{111}$$

  where κ is the thermal conductivity of the material.

- **Convection:** Transfer of heat by the bulk motion of a fluid (e.g., hot air rising).
- **Radiation:** Transfer of energy via electromagnetic waves. This requires no medium (e.g., sunlight).

93 The Nature of Heat

93.1 The Mechanical Equivalent of Heat

Historically, heat was thought to be a fluid ("caloric"). James Joule proved that mechanical work can generate heat. By churning water with a paddle wheel driven by

falling weights, he showed that the loss of mechanical potential energy (mgh) resulted in a proportional rise in water temperature.

$$W \propto Q \implies 1 \text{ cal} = 4.184 \text{ Joules} \tag{112}$$

This established the **First Law of Thermodynamics**: Energy is conserved, and heat is simply the transfer of energy.

93.2 The Microscopic View

- **Kinetic Energy:** Organized motion of atoms (e.g., a thrown baseball).
- **Heat (Thermal Energy):** Disorganized, random motion of atoms.

When a car brakes, its organized kinetic energy is not lost; it is converted into the random vibrational energy of the atoms in the brake pads (heat).

94 The Ideal Gas Law and Microscopic Physics

Empirically, dilute gases obey a simple relationship between Pressure (P), Volume (V), and Temperature (T).

94.1 The Equation of State

For an ideal gas, the product of pressure and volume is proportional to the absolute temperature and the amount of gas.

$$PV = NkT \tag{113}$$

where:

- N: The total number of molecules.
- k: The **Boltzmann Constant** ($k \approx 1.38 \times 10^{-23}$ J/K).

This form highlights that pressure depends on the number of particles, not their mass.

94.2 Moles and the Gas Constant

Because N is typically huge ($\sim 10^{23}$), we often count particles in **moles**.

- **Avogadro's Number (N_A):** The number of particles in one mole.

$$N_A \approx 6.022 \times 10^{23} \text{ particles/mol}$$

- The number of moles n is given by $n = N/N_A$.

Substituting $N = nN_A$ into the ideal gas law:

$$PV = n(N_A k)T$$

We define the **Universal Gas Constant** R:

$$R = N_A k \approx 8.31 \, \text{J}/(\text{mol} \cdot \text{K}) \tag{114}$$

Thus, the macroscopic form of the ideal gas law is:

$$PV = nRT \tag{115}$$

94.3 Physical Interpretation of Temperature

The dimensions of Pressure × Volume are Force/Area × Volume = Force × Distance = **Energy**. Since $PV = NkT$, the term kT must have dimensions of energy. This suggests that **Temperature is a measure of the average energy per molecule**. Specifically, for an ideal monatomic gas:

$$\text{Average Kinetic Energy} = \frac{3}{2}kT$$

This connects the macroscopic sensation of "hotness" directly to the microscopic speed of atoms.

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96 Kinetic Theory of Gases

We can derive the macroscopic Ideal Gas Law ($PV = NkT$) by analyzing the microscopic collisions of atoms with the container walls.

96.1 Derivation of Pressure

Consider a cubic box of side L containing N particles of mass m.

- A particle moving with velocity v_x hits the wall and rebounds with velocity $-v_x$.
- Momentum change: $\Delta p = 2mv_x$.
- Time between collisions with the same wall: $\Delta t = \frac{2L}{v_x}$.
- Average force exerted by one particle: $F = \frac{\Delta p}{\Delta t} = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L}$.

Summing over all N particles:

$$F_{\text{total}} = \frac{m}{L}\sum v_{x,i}^2 = \frac{Nm\langle v_x^2\rangle}{L}$$

Since motion is random, $\langle v_x^2\rangle = \langle v_y^2\rangle = \langle v_z^2\rangle = \frac{1}{3}\langle v^2\rangle$. The Pressure is $P = F_{\text{total}}/L^2$:

$$P = \frac{Nm\langle v^2\rangle}{3L^3} = \frac{Nm\langle v^2\rangle}{3V}$$

Rearranging gives:

$$PV = \frac{2}{3}N\left(\frac{1}{2}m\langle v^2\rangle\right) \tag{119}$$

96.2 Microscopic Definition of Temperature

Comparing the kinetic theory result with the Ideal Gas Law ($PV = NkT$):

$$\frac{2}{3}N\langle K\rangle = NkT$$

$$\langle K\rangle = \frac{3}{2}kT \tag{120}$$

This is the fundamental definition of temperature in classical physics: **Temperature is proportional to the average translational kinetic energy of the molecules.**

- At $T = 0$ K (Absolute Zero), all molecular motion ceases.
- Heavy and light atoms at the same temperature have the same kinetic energy (so lighter atoms move faster).

96.3 Thermodynamic Processes

The state of a gas is defined by (P, V, T). Since $PV = nRT$, specifying two variables fixes the third. We visualize states as points on a **PV Diagram**.

- **Equilibrium:** If the system is static, it has a well-defined point (P, V).
- **Non-Equilibrium:** If a piston is moved rapidly, turbulence occurs. Pressure is not uniform. The system does not have a single point on the diagram.
- **Quasi-Static Process:** If changes are made infinitely slowly, the system remains in equilibrium throughout. This traces a continuous path (curve) on the PV diagram.

97 The First Law of Thermodynamics

We define the **Internal Energy (U)** of an ideal gas as the sum of the kinetic energies of all its molecules. Since $\langle K \rangle \propto T$, the internal energy depends **only** on the temperature:

$$U = \frac{3}{2} nRT \tag{121}$$

If the temperature is constant (Isothermal), the internal energy does not change ($\Delta U = 0$).

97.1 Statement of the First Law

The First Law is the conservation of energy for thermodynamic systems. The change in internal energy equals the heat added minus the work done by the system.

$$\Delta U = Q - W \tag{122}$$

- $Q > 0$: Heat added to the system.
- $W > 0$: Work done **by** the system (expansion).

97.2 Work Done by a Gas

Work is defined as Force times distance. For a gas pushing a piston of area A by a distance dx:

$$dW = Fdx = (PA)dx = P(Adx) = PdV$$

The total work done in a quasi-static process is the integral:

$$W = \int_{V_1}^{V_2} P(V)\, dV \tag{123}$$

Geometrically, this is the **Area under the curve** on a PV diagram.

Example: Isothermal Expansion
For an ideal gas at constant temperature T, $P = nRT/V$.

$$W = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln\left(\frac{V_2}{V_1}\right)$$

Since $\Delta T = 0$, $\Delta U = 0$. By the First Law, $Q = W$. All the heat added is converted directly into work.

97.3 Molar Specific Heats

For gases, the amount of heat required to raise the temperature depends on the process. We define specific heat per mole (C).

1. **Constant Volume (C_V):** The volume is fixed, so $W = \int PdV = 0$. All heat goes into internal energy.

$$Q = \Delta U = \frac{3}{2}nR\Delta T \implies C_V = \frac{3}{2}R$$

2. **Constant Pressure (C_P):** The gas expands at constant pressure, doing work $W = P\Delta V = nR\Delta T$. The heat required is:

$$Q = \Delta U + W = \frac{3}{2}nR\Delta T + nR\Delta T = \frac{5}{2}nR\Delta T$$

$$C_P = \frac{5}{2}R$$

Mayer's Relation: For any ideal gas:

$$C_P = C_V + R \tag{124}$$

The ratio of specific heats is called the adiabatic index γ:

$$\gamma = \frac{C_P}{C_V}$$

For a monatomic ideal gas, $\gamma = 5/3 \approx 1.67$.

98 Thermodynamic Processes and Cycles

We apply the First Law ($\Delta U = Q - W$) to specific quasi-static processes for an ideal gas. Recall that for an ideal monatomic gas, $U = \frac{3}{2}nRT$.

98.1 Types of Processes

1. **Isochoric (V is constant):**
   - Work: $W = \int PdV = 0$.
   - First Law: $\Delta U = Q$.
   - Specific Heat: $C_V = \frac{3}{2}R$.
2. **Isobaric (P is constant):**

- Work: $W = P\Delta V = nR\Delta T$.
- First Law: $\Delta U = Q - P\Delta V$.
- Specific Heat: $C_P = C_V + R = \frac{5}{2}R$.

3. **Isothermal (T is constant):**
   - Change in Energy: $\Delta U \propto \Delta T = 0$.
   - First Law: $Q = W$.
   - Work: $W = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln(V_2/V_1)$.
   - Physics: The gas expands and does work, but absorbs exactly enough heat from the reservoir to keep its temperature from dropping.
4. **Adiabatic ($Q = 0$):**
   - No heat enters or leaves (insulated system).
   - First Law: $\Delta U = -W$.
   - Physics: If the gas expands ($W > 0$), internal energy decreases ($\Delta U < 0$), so the gas cools down.
   - Relation: $PV^\gamma = \text{constant}$, where $\gamma = C_P/C_V$.

98.2 Thermodynamic Cycles

A cycle is a process that returns the system to its initial state (e.g., A $\to$ B $\to$ C $\to$ A).

- **State Variables (ΔU):** Since the system returns to the start, the total change in internal energy is zero. $\oint dU = 0$.
- **Path Variables (Q, W):** The net work done is the area enclosed by the loop in the PV diagram.
- **First Law for a Cycle:**

$$\Delta U_{\text{cycle}} = 0 \implies Q_{\text{net}} = W_{\text{net}}$$

This is the principle behind heat engines: we take heat from a source, convert some of it to work, and dump the rest, returning the engine to its initial state to repeat the cycle.

99 Adiabatic Processes

An adiabatic process is one where no heat is exchanged with the environment ($Q = 0$). This occurs in insulated systems or rapid processes (like sound waves or the power stroke of an engine).

99.1 Derivation of the Adiabatic Law

From the First Law with $Q = 0$:

$$dU = -dW \implies nC_V dT = -PdV$$

From the Ideal Gas Law ($PV = nRT$), differentiating gives:

$$PdV + VdP = nRdT$$

Solving for dT and substituting into the energy equation:

$$nC_V \left(\frac{PdV + VdP}{nR} \right) = -PdV$$

Using $R = C_P - C_V$ and rearranging terms leads to:

$$C_V VdP + C_P PdV = 0$$

Dividing by PVC_V:

$$\frac{dP}{P} + \frac{C_P}{C_V} \frac{dV}{V} = 0$$

Let $\gamma = C_P/C_V$. Integrating gives $\ln P + \gamma \ln V = \text{constant}$, which implies:

$$PV^\gamma = \text{constant} \tag{125}$$

Alternatively, using $P \propto T/V$, we can write:

$$TV^{\gamma-1} = \text{constant}$$

99.2 Work Done in an Adiabatic Process

Since $P = KV^{-\gamma}$ (where K is a constant), the work done is:

$$W = \int_{V_1}^{V_2} KV^{-\gamma} dV = K \left[\frac{V^{1-\gamma}}{1-\gamma} \right]_{V_1}^{V_2}$$

Substituting $K = P_1 V_1^\gamma = P_2 V_2^\gamma$ back into the limits:

$$W = \frac{P_2 V_2^\gamma V_2^{1-\gamma} - P_1 V_1^\gamma V_1^{1-\gamma}}{1-\gamma} = \frac{P_2 V_2 - P_1 V_1}{1-\gamma}$$

Rewriting to make the denominator positive:

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1} \tag{126}$$

This formula calculates the work done solely from the change in the state of the gas, without needing to calculate the integral explicitly.

100 The Second Law of Thermodynamics

While the First Law (Conservation of Energy) tells us that energy cannot be created or destroyed, it does not forbid processes like a shattered egg reassembling itself. The Second Law explains the direction of natural processes.

100.1 Irreversibility and Entropy

Natural processes tend to move from states of order to states of disorder. We quantify this disorder with a variable called **Entropy (S)**. **The Second Law:** The total entropy of an isolated system (or the universe) never decreases.

$$\Delta S_{\text{universe}} \geq 0$$

This explains why heat flows from hot to cold: the mixed state has higher entropy (more disorder) than the separated state.

100.2 Heat Engines

A heat engine is a device that absorbs heat Q_H from a hot reservoir at T_H, performs work W, and expels waste heat Q_C to a cold reservoir at T_C.

- **First Law applied to Engine:** $Q_H = W + Q_C$.
- **Efficiency (η):** Defined as what you get (Work) divided by what you pay for (Heat Input).

$$\eta = \frac{W}{Q_H} = \frac{Q_H - Q_C}{Q_H} = 1 - \frac{Q_C}{Q_H} \tag{127}$$

100.3 The Carnot Cycle

Sadi Carnot devised a theoretical engine that operates reversibly between two temperatures. It consists of four steps:

1. Isothermal Expansion at T_H (Heat Q_H in).
2. Adiabatic Expansion (Temp drops to T_C).
3. Isothermal Compression at T_C (Heat Q_C out).
4. Adiabatic Compression (Temp rises to T_H).

For this ideal cycle, it can be proven that the ratio of heat exchanged matches the ratio of temperatures:

$$\frac{Q_C}{Q_H} = \frac{T_C}{T_H}$$

Substituting this into the efficiency formula gives the **Carnot Efficiency**:

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H} \tag{128}$$

This represents the fundamental physical limit on the efficiency of any heat engine operating between two temperatures. No real engine can exceed this efficiency.

101 The Adiabatic Process

An **Adiabatic Process** is one where the system is thermally isolated from its environment.

$$\Delta Q = 0 \tag{129}$$

This occurs when the system is heavily insulated or when the process occurs too rapidly for heat to flow (e.g., sound waves).

101.1 Physical Intuition

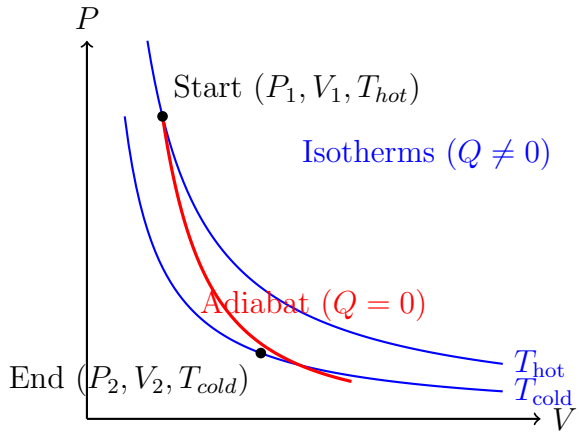
Consider a gas expanding pushing a piston:

- **Work:** The gas does positive work ($W > 0$).
- **First Law:** Since $Q = 0$, we have $\Delta U = -W$.
- **Result:** The internal energy decreases, which means the **Temperature drops**.

Because the temperature drops during expansion, the pressure drops faster than it would in an isothermal process. The adiabatic curve on a PV diagram is **steeper** than the isotherms.

101.2 PV Diagram: Isotherms vs. Adiabats

In the figure below, the blue curves are Isotherms (constant T). The red curve is an Adiabat. Notice how the gas starts at a high temperature (T_2), expands, and drops to a lower temperature (T_1).



Comparison of Isothermal and Adiabatic expansion. The Adiabatic path crosses from a higher temperature isotherm to a lower one.

101.3 The Equations of State

For an ideal gas undergoing a reversible adiabatic process, the variables are related by:

$$PV^{\gamma} = \text{constant} \tag{130}$$

$$TV^{\gamma-1} = \text{constant} \tag{131}$$

where $\gamma = C_P/C_V$ is the adiabatic index (ratio of specific heats). For a monatomic gas, $\gamma = 5/3$.

101.4 Work Done in Adiabatic Expansion

Since heat is zero, the work done comes entirely from the loss of internal energy.

$$W = -\Delta U = nC_V(T_1 - T_2)$$

Alternatively, in terms of pressure and volume:

$$W = \frac{P_1V_1 - P_2V_2}{\gamma - 1} \tag{132}$$

102 Entropy and Statistical Mechanics

We have seen that entropy increases in irreversible processes (like heat flow or free expansion). But what is entropy physically? Ludwig Boltzmann provided the answer by connecting thermodynamics to statistics.

102.1 Calculating Entropy Changes

Entropy is a state function. To calculate the change ΔS between state A and state B, we must choose a **reversible path** connecting them, even if the actual process was irreversible.

$$\Delta S = \int_A^B \frac{dQ_{\text{rev}}}{T} \tag{133}$$

Example: Free Expansion
A gas of volume V expands into a vacuum to volume $2V$. No work is done, no heat flows ($Q = 0, W = 0$). The temperature stays constant.

- **Wrong Way:** $\Delta S = \int 0/T = 0$. (Incorrect because free expansion is irreversible).
- **Right Way:** Design a reversible process. Imagine a piston slowly expanding the gas from V to $2V$ while keeping T constant (Isothermal Expansion).

$$Q_{\text{rev}} = W = \int_V^{2V} \frac{nRT}{V} dV = nRT \ln 2$$

$$\Delta S = \frac{Q_{\text{rev}}}{T} = nR \ln 2$$

The entropy increases, reflecting the irreversibility of the expansion.

102.2 The Statistical Definition of Entropy

Boltzmann realized that macroscopic states (like "gas fills the box") can be formed by many different microscopic arrangements of atoms.

- **Microstate:** A specific specification of the position and velocity of every single atom.
- **Multiplicity (Ω):** The number of different microstates that correspond to the same macroscopic state.

Boltzmann's Formula for Entropy is:

$$S = k \ln \Omega \tag{134}$$

where k is the Boltzmann constant.

102.3 The Arrow of Time

Why does entropy always increase? Consider a box with N molecules.

- **Ordered State:** All molecules in the left half. $\Omega = 1$ (roughly). $S \approx 0$.
- **Disordered State:** Molecules spread evenly. $\Omega \propto 2^N$ (huge number). S is large.

If the system evolves randomly, it will inevitably move toward the state with the highest multiplicity Ω. The Second Law is a statistical law: Systems evolve toward the most probable state (highest entropy). Events like a shattered egg reassembling or air gathering in one corner of a room are not impossible, but they are statistically so improbable ($\sim 1/2^{10^{23}}$) that they never happen in the age of the universe.